

AN ARCHIVAL ALMA SEARCH FOR UNRESOLVED SPECTRAL-CARRIER TECHNOSIGNATURES
TOWARD STARS WITHIN 40 PC AT 90–873 GHzGLENN J. WHITE^{1,2} AND ROBIN DEY³

ABSTRACT

The millimetre and submillimetre sky has never been searched for narrowband transmitters toward nearby stars. Public ALMA observations taken for other purposes record a spectrum of the star at each field centre, so we searched them: 402 spectral windows toward 60 archive-selected stellar systems within 40 pc, channels fine enough to follow a drift, covering 47.7 GHz of union bandwidth in disconnected intervals between 114 and 873 GHz. Of the 56 threshold crossings, 20 coincide with a catalogued molecular transition — among them carbon monoxide in the β Pictoris disc, which the search recovered without being told where to look — and the remainder are no more numerous than chance predicts. Each of the 36 unattributed crossings a later observation covered was tested for recurrence at its predicted stellar-frame frequency and drift, and no event satisfied the pre-defined independent-recurrence confirmation criterion. The null has a limit: only 30 of the 60 systems were observed again at the same frequency more than a day apart, over 23.3 GHz of the union searched, so the searched extent is about twice the confirmable one. Depth is measured by injecting carriers into the calibrated visibilities and recovering them: the median system’s best window returns nine in ten at 1.6×10^{15} W, a recovery fraction and not a detection probability. The same data’s coarser windows limit an unresolved spectral excess. For classifying a crossing in the present archival data set, molecular-line confusion and sparse repeat coverage are more immediate limitations than thermal sensitivity.

Subject headings: extraterrestrial intelligence, radio lines: stars, submillimetre: stars, surveys, astrobiology

1. INTRODUCTION

The public ALMA archive holds thousands of hours of millimetre and submillimetre observations taken for other reasons, mostly studies of dust discs and planet formation. Their visibilities allow a spectrum at the position of the star at the centre of each field, at a resolution and depth set by the original observer’s correlator setup. This paper searches those spectra for a narrowband transmitter at the star’s own position. Nothing of the kind survived.

Searches for *technosignatures* — detectable imprints of technology on the electromagnetic spectrum, intended or not (Sheikh 2020; Vidal et al. 2026) — have examined many thousands of stars over six decades, yet sample only 6×10^{-18} of the space of transmitter frequency, bandwidth, power and duty cycle (Wright et al. 2018), and essentially all of that effort lies below ~ 30 GHz. No published radio technosignature search above ~ 30 GHz has reached a star within 40 pc: the one published search in that range worked in ALMA Band 3 toward targets at 1.01 kpc and beyond (Mason et al. 2025), and few searches at any frequency reprocess existing interferometric archives (White et al. 2026). The band is attractive because a carrier arrives less distorted and a given aperture delivers more gain per watt supplied, and hard because archived correlator channels are thousands of times coarser than a dedicated narrowband backend and

are crowded with molecular lines, a dense background of real astrophysical signals that a threshold search must separate from an artificial one (§2, §6.1).

The signal sought is an *unresolved spectral carrier*, emission confined to a few channels, unresolved at the stellar position, and persistent enough to recur between epochs. Each qualifier is a restriction. A carrier broader than the channel width would not be recovered, nor one that fills the synthesised beam, nor one that appears once and never again — and it is recurrence rather than brightness that separates an interesting event from a detection (Sheikh et al. 2021). The frequency is allowed to drift, as it must for a transmitter carried on a rotating, orbiting body, and the drift searched is bounded by a line-of-sight acceleration rather than by a frequency (§4.1). The channels used are 15.3–1953 kHz wide, fine only relative to this archival sample, so “narrowband” here describes the assumed transmitter and never the resolving power of the data. The targets were chosen by archival disc and planet-formation programmes and not for this search, so every statistical statement that follows is about an archive-selected sample and not about the solar neighbourhood (§6.3).

2. BACKGROUND

Radio searches for technosignatures began at centimetre wavelengths and have largely stayed there. The first targeted experiments (Cocconi & Morrison 1959; Drake 1961), the programmes reviewed by Tarter (2001) and Project Phoenix (Backus & Project Phoenix Team 2002) all worked below a few gigahertz. The modern generation divides into targeted and commensal surveys at centimetre and metre wavelengths (Zhang et al. 2020; Czech et

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al. 2021; Morrison et al. 2023; Li et al. 2026; Johnson et al. 2023), with wide-field low-frequency coverage from the MWA (Tremblay et al. 2024) and single-dish C- and K-band work at 6 and 18 GHz with the Sardinia Radio Telescope (Manunza et al. 2025). Machine-learning interference rejection is now central to that effort (Ma et al. 2023; Choza et al. 2024; Jacobson-Bell et al. 2025; Poznanski et al. 2025; Garrett et al. 2026). The classical threshold search used here lacks that capability, and instead uses a spatial test on the star against control positions in the same field, which a single dish cannot make.

Above about 30 GHz the record is nearly empty, and that is what makes this frequency range new. ALMA observes in Bands 3 to 10, spanning roughly 84 to 950 GHz; of those eight bands, one has been searched for technosignatures. For continuity with the centimetre literature we note one familiar power scale: the Arecibo planetary radar radiated 2.0×10^{13} W at 2.38 GHz (Drake 1974; Sheikh et al. 2025). It carries no transmitter model into the millimetre band, where an aperture of that diameter is not realisable, so the illustrative benchmark of §6.3 is instead a dish of 12 m. Neither is a privileged scale; the limits of this paper are stated as EIRP and need no transmitter model.

Mason et al. (2025) is the only prior millimetre search and the only comparable archival one. They searched Band 3 windows toward 28 Gaia DR3 stars lying in the fields of 4 ALMA calibrators, observed for other science and so usable for free. Their nearest target is at 1.01 kpc, where a 5σ flux threshold corresponds to $\text{EIRP}_{\min} > 6.9 \times 10^{17}$ W. That target lies $25\times$ further away than the most distant star searched here and $277\times$ further than the nearest: the two samples differ by 1.4–2.4 orders of magnitude in distance, and at a fixed received flux by the square of that, which is 3–5 orders in minimum detectable EIRP. An EIRP ratio between the two surveys would therefore be almost entirely the d^2 term. The quantity that does not depend on distance is the minimum detectable *received* flux, $F_{\min} = S_{\min} \delta\nu$. It must be quoted at a matched channel width, because $S_{\min}$ falls as $\delta\nu^{-1/2}$ while $F_{\min}$ rises as $\delta\nu^{1/2}$. Mason et al. reach 3.8×10^{-23} – 5.8×10^{-23} W m $^{-2}$ across their four fields, on a 30.52 kHz channel at a bare 5σ flux threshold. On that same channel width and the same threshold convention, the fine-channel windows searched here reach 5.2×10^{-24} W m $^{-2}$ at best, a factor 9 deeper, and 7.1×10^{-23} W m $^{-2}$ at the median, a factor 1.5 shallower. The two experiments are therefore comparable in depth and differ in what they cover: distant stars in four calibrator fields, against 60 stellar systems within 40 pc searched for a drifting carrier, over a frequency range reaching about nine times higher. They also identified the obstacles that shape the present method: drift rates scale with frequency, smearing erodes sensitivity unless drift is searched for, and the dense forest of molecular lines raises the confusion background. Every window reported here was re-reduced and re-searched from the raw archive data with the pipeline of §4, so for any star in common the numbers to cite are this paper’s.

Figure 1(a) places this survey among published searches, in minimum detectable EIRP against distance. The vertical axis is physically commensurable: for a car-

rier narrower than every programme’s channel, the minimum detectable EIRP is a total radiated power and not a spectral power density, so the same quantity is plotted for each programme. What differs is the convention, so both are drawn. Every published limit shown is a nominal 5σ threshold at which recovery is not stated, and this survey’s filled symbols are the same thing measured the same way; the open symbols above them are EIRP_{90} , a 90 per cent recovery through the complete automated selection, which is the stricter statement and the one every limit in this paper is quoted on. Any EIRP advantage toward the nearest systems is a d^2 effect and not an instrumental one. Panel (b) shows the other axis that matters. Archived correlator channels here span about seven orders of magnitude in width, and all of them are far coarser than a dedicated narrowband backend. The distinctive feature of this survey is therefore its frequency regime and its channelisation, not its depth. It also samples far fewer systems and far fewer epochs than a dedicated search, and complements those programmes rather than deepening them.

A non-detection becomes a quantitative statement only through a stated search volume. Wright et al. (2018) formalised that volume, and Margot et al. (2023) its conversion into an upper bound on the fraction of stars hosting a transmitter; Sheikh et al. (2025) asked where Earth’s own technosignatures would be detectable. Such a bound is valid only if the sample represents the population it is meant to describe. This sample was assembled by other observers for other purposes, so it does not, and §3 states what it is instead.

3. THE SAMPLE

This search covers 1651 spectral windows toward 89 stars in 82 systems within 40 pc, drawn from 404 public ALMA execution blocks and 4.1 TB of raw data through one frozen pipeline, 1094 h on source. We call that set the *primary census*, and it is the set every limit, every crossing and every expectation in this paper is computed over. Four further sets of blocks enter the paper — a hold-out reserved before the archive was searched, an archival extension searched afterwards, an out-of-sample control beyond 40 pc, and an external sample used only to calibrate the spatial statistic — and none of them is part of the primary census. The primary census is 404 execution blocks, 1651 searched windows, 89 stars in 82 systems and 1094 h on source; the hold-out is 77 blocks and 315 windows, and the archival extension 115 blocks and 463 windows. Table 7 gives all five sets in rows that sum, with the results each one enters; wherever a number in this paper is computed over something other than the primary census, that table’s name for it is used at the point of use.

Three words are used consistently and are not interchangeable. A *star* is an individual stellar component; a *system* is a gravitationally bound group, counted once however many of its components were observed; a *target* is one ALMA pointing. The census holds 89 stars in 82 systems.

The census is two experiments, not one. 402 windows toward 65 stars in 60 systems are fine-channel, called *Class A* throughout: these resolve a drifting carrier and are the primary experiment, and every sensitivity and every exclusion quoted here refers to them un-

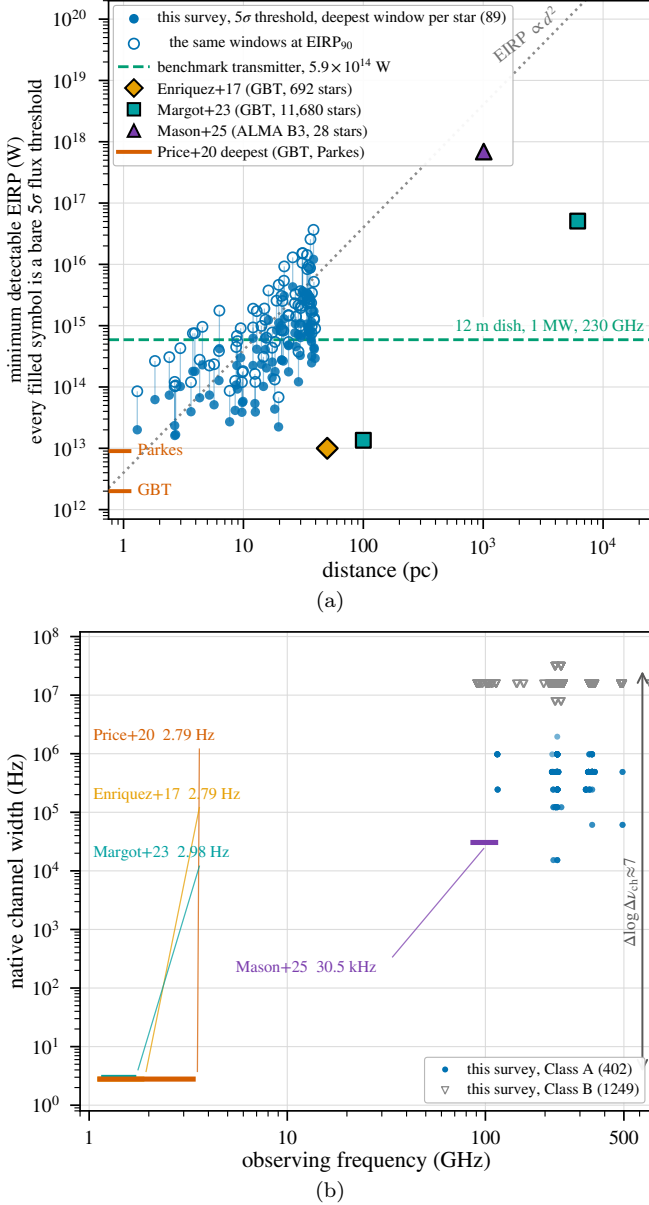


FIG. 1.— Where this survey sits among published radio technosignature searches. (a) Minimum detectable EIRP against distance. Blue, this survey, the deepest window per star: *filled*, that window’s bare 5σ threshold, which is the quantity the other programmes quote; *open*, the same window at EIRP₉₀, joined to its own filled symbol. Orange, Enriquez et al. (2017); aqua, Margot et al. (2023); violet, Mason et al. (2025); left-edge ticks, Price et al. (2020), all at their own 5σ thresholds. Grey dotted, $\text{EIRP} \propto d^2$ at constant received flux; green dashed, the illustrative 12 m benchmark transmitter of §6.3. The ordinates are commensurable as total powers for a sub-channel carrier, and the filled symbols make the comparison on one convention; the gap to the open symbol is what completeness costs. (b) Native channel width against observing frequency for every searched window (filled, Class A; open, Class B), with the comparison programmes as horizontal segments. The channel widths span about seven orders of magnitude, which is what the $\Delta \log \Delta \nu_{\text{ch}} \approx 7$ annotation marks: the channelisation, not the plotted power, is what differs most between programmes.

less stated otherwise. The remaining 1249 windows are coarse-channel, *Class B*, and support a secondary search for unresolved spectral excess; they can register a spike but cannot distinguish a drifting signal from a stationary one.

The inclusion criteria are objective and were applied by query. A star enters if its Gaia DR3 parallax is at least 25 mas, measured to better than 20 per cent, and if its proper-motion-corrected position at the observation epoch falls inside the quoted field of view of at least one public execution block. Distances are direct inversions, $d = 1/\varpi$, and the star need not sit at the pointing centre. Disc, planet and spectral type enter as descriptive meta-data only, and no star or window was excluded by an unlisted criterion or for short integration. The parallax criterion never binds: the worst fractional parallax error among the stars searched is 2.5 per cent, an order of magnitude inside it, and inside the stricter $\sigma_\varpi/\varpi \leq 0.1$ used in Table 1 to select a reference census on the same footing. The query returns 168 work-list entries; the quoted field of view is optimistic, and only 115 of those stars are in fact contained by a public field, of which 89 yielded a searchable window. The per-star and per-window ledger, including the five-criterion decision tree and every edge case, is in the data release.

3.1. The archival selection

This sample was not designed. It is whatever previous ALMA programmes happened to observe, for their own reasons, and 90 per cent of the stars searched were targeted under a debris-disc or planet-formation proposal. The resulting bias is strong and measurable, and Table 1 tabulates it against a volume-limited reference census selected on the same parallax quality.

The dominant selection effect is distance. Stars within 5 pc are over-represented by a factor 49, and the enhancement falls monotonically with distance to a factor of two *under*-representation beyond 30 pc. Spectral type comes second: A stars are over-represented by about 23 and F and G stars by factors of two to three, because those are the stars with bright debris discs. M dwarfs are mildly depleted — 41 of the 89 searched stars are M dwarfs, a factor 0.8 of their share of the reference census, and only 29 of them carry the fine-channel coverage that can resolve a drifting carrier at all — and that is a weaker statement than the table’s early-type ratios, not a stronger one. Both figures are bounds in the same direction: Gaia temperatures are missing for about half the reference census and missing preferentially for the coolest objects, so the census M fraction is a lower limit and every early-type ratio is therefore a lower limit too. Coverage is also concentrated: 285 of the 1651 windows come from 3 systems. Every statistical statement in this paper is therefore a statement about an archive-selected sample of nearby stars, and none of them is an occurrence rate for the solar neighbourhood. The systems a targeted search would itself have chosen — every sample system within 15 pc with a confirmed planet — are listed in the data release, where 6 of the 15 carry no Class A coverage at all and so no drift-resolving limit: the archival selection acting on exactly the targets a transmitter search would pick.

3.2. Frequency coverage and archival scope

TABLE 1

SELECTION FUNCTION OF THE SEARCHED SAMPLE AGAINST A VOLUME-LIMITED REFERENCE CENSUS, BOTH SELECTED ON THE SAME PARALLAX QUALITY: GAIA DR3 SOURCES WITH $\varpi > 0$, $\sigma_{\varpi}/\varpi \leq 0.1$ AND $1/\varpi \leq 40$ pc (17 566 OBJECTS, 8594 OF THEM CARRYING A `TEFF_GSPPHOT` ESTIMATE), AGAINST THE 89 STARS WITH AT LEAST ONE SEARCHED WINDOW. THE SEARCHED STARS ARE MEASURED FAR BETTER THAN THE CUT REQUIRES — THE WORST FRACTIONAL PARALLAX ERROR AMONG THEM IS 2.5 PER CENT — SO THE TWO COLUMNS ARE SELECTED ALIKE, AND LOOSENING THE CUT TO 0.2 ADDS NO CENSUS OBJECT AT ALL. 86 OF THE 89 SEARCHED STARS ARE CLASSIFIED: 53 FROM THE GAIA TEMPERATURE, 25 FROM THE SIMBAD SPECTRAL TYPE WHERE THAT TEMPERATURE IS ABSENT, AND 8 IDENTIFIED AS WHITE DWARFS; 3 CARRY NO CLASSIFICATION IN EITHER SOURCE.

THE SIMBAD ROUTE MATTERS BECAUSE `TEFF_GSPPHOT` IS MISSING PREFERENTIALLY FOR COOL STARS, WHICH IS THE POPULATION THE M DWARF STATEMENT IS ABOUT. “RATIO” IS THE SAMPLE COLUMN FRACTION OVER THE CENSUS COLUMN FRACTION; > 1 IS OVER-REPRESENTATION. *The strongest selection effect here is distance, not spectral type.*

Property	Census	Searched	Ratio
Distance			
$d \leq 5$ pc	48 (0.3%)	12 (13.5%)	49.3
5–10 pc	240 (1.4%)	13 (14.6%)	10.7
10–20 pc	2069 (11.8%)	18 (20.2%)	1.7
20–30 pc	5391 (30.7%)	19 (21.3%)	0.7
30–40 pc	9818 (55.9%)	27 (30.3%)	0.5
T_{eff} class ^a			
A	24 (0.3%)	5 (6.4%)	23.0
F	402 (4.7%)	9 (11.5%)	2.5
G	860 (10.0%)	16 (20.5%)	2.0
K	1400 (16.3%)	7 (9.0%)	0.6
M	5908 (68.7%)	41 (52.6%)	0.8
Other axes			
Confirmed planet hosts ^b	26 (15.5%)	23 (25.8%)	1.7
White dwarfs ^a	—	8 (9.0%)	—
Searched stars / systems	—	89 / 82	—

^aTemperature bins, not MK types: A ≥ 7500 , F 6000–7500, G 5300–6000, K 3900–5300, M < 3900 K, normalised over the 78 classified non-degenerate stars. The 8 white dwarfs are kept apart because a degenerate is not an early-type star and its photometric temperature would place it among them. Gaia `teff_gspphot` is unavailable for about half the reference census, preferentially for the coolest and faintest objects, so the census M fraction is a lower limit and every earlier-type ratio in this panel is correspondingly a lower limit; the searched column has no such gap.

^bBoth columns are the same positional join, of the 168-entry ALMA-covered work list and of its searched subset, to the NASA Exoplanet Archive `pscomppars` table accessed 2026 October 4; a confirmed planet is a row of that table. The 23 searched hosts carry 50 confirmed planets between them, and 21 of the 23 survive the deuterium-burning mass limit. The Gaia reference census carries no planet information, so the census column in this row is the work list rather than the 17 566-object Gaia census.

The Class A experiment covers $\Delta\nu_A = 47.7$ GHz of unique sky frequency, and that coverage is sparse: it is 26 disjoint islands scattered between 114.0 and 873.1 GHz, not a band. The endpoints are a range over which observations exist, never a range that was covered. Adding the coarse-channel windows brings the total archival coverage to 118.1 GHz in 46 intervals, of which 113.3 GHz survives the molecular-line mask of §4. The distribution is shown in Fig. 2; a transmitter at a frequency between the islands was never observable here, whatever its power.

The archive is not exhausted either, and the shortfall is accounted for rather than estimated. 177 further blocks lie inside the same query but were not processed when the census was frozen; Table 7 gives what became of each. 144 of them have since been calibrated and searched, and they enter this paper only as repeat epochs. The largest

single gap is the 18 blocks for which the archive holds raw visibilities that were never pipeline-calibrated, 237 GB in all; no effort on our side would close it.

4. DATA AND METHODOLOGY

The search looks for one kind of signal: an unresolved carrier at the star’s own position, narrow compared with a single correlator channel, and drifting in sky frequency because the transmitter accelerates along the line of sight. Over a long track its power is spread across many channels, so a search that treats each channel on its own will miss it. The analysis de-drifts the data before it measures them.

4.1. The de-drifted search statistic

Each window’s dynamic spectrum is stacked along every trial drift rate $\dot{\nu}$. The search statistic is the largest stacked signal-to-noise ratio anywhere on the resulting channel $\times$ drift grid,

$$T(\mathbf{x}) = \max_{\dot{\nu}, \nu} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot{\nu}t) / \sigma^2(t, \nu)}{[\sum_t 1 / \sigma^2(t, \nu)]^{1/2}}, \quad (1)$$

where $I(\mathbf{x}, t, \nu)$ is the spectrum at sky position $\mathbf{x}$ in integration t and channel ν , and $\sigma(t, \nu)$ is the noise scale there. Equation 1 is the matched filter for a constant-drift carrier: it adds the signal coherently along the track and divides by the noise of the stacked spectrum, so T is an absolute signal-to-noise ratio and not a local contrast. The noise scale is the median absolute deviation across 512 control positions at each integration and each channel, scaled to its Gaussian-equivalent standard deviation — local in time and frequency, global in position, shared by the star and every control, and with the across-position median never subtracted from the numerator. The statistic is evaluated at one position at a time, identically at the star and at each control, which is what makes the two comparable. T_* denotes its value at the star.

Spectra are extracted at the star’s apparent position, never from images. For each integration and each native channel the pipeline forms the weighted vector average of the phase-shifted visibilities, in total intensity: the mean $I = (XX + YY)/2$ of the two parallel-hand correlations. That average is the dirty-map value at the extraction position — an amplitude, not a deconvolved flux density, so in a field with extended emission it is not a point-source measurement. The data are used as the archive delivered them, through ALMA’s own calibration and quality assurance, with their flagging carried through, every delivered channel searched, and no regridding, deconvolution or edge trimming. A per-integration median along the *frequency* axis removes the continuum; no median is taken along time, so a one- or two-channel feature cannot be removed by it. One subtlety matters for every power quoted below: the delivered channels are smoothed, so a channel’s effective noise bandwidth is not its separation and flux density times channel width is not yet a radiated power. A response correction computed from the correlator’s own channel shape supplies the difference, derived and validated against the effective resolutions ALMA reports in Appendix A.

One convention is fixed here and holds everywhere in the paper. The search runs on the topocentric sky fre-

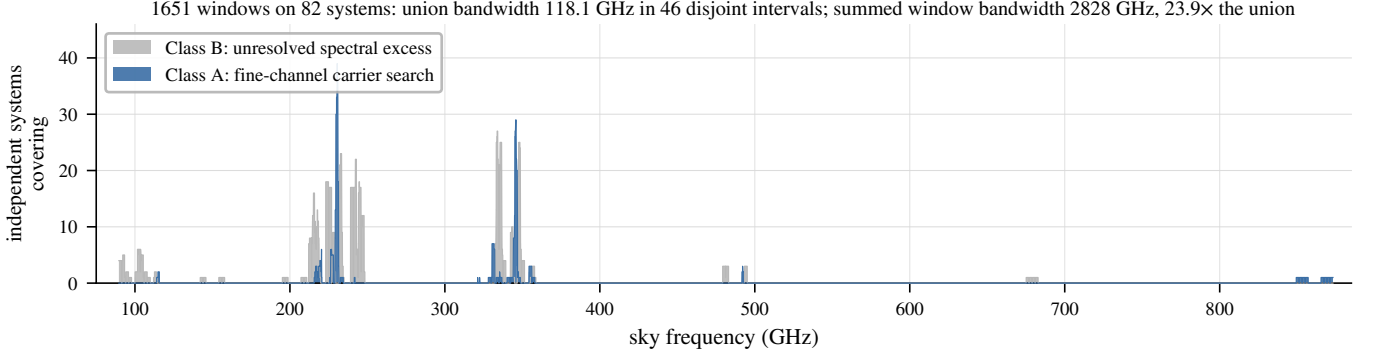


FIG. 2.— Where this survey searched. *Lower panel*: each searched window is a horizontal segment on its host system’s row, rows ordered by distance and labelled alternately on the left and right axes; heavy blue marks the fine-channel carrier search (Class A) and light grey the coarse-channel spectral-excess search (Class B). *Upper panel*: the number of systems covering each gigahertz of sky frequency; the threshold distribution is Fig. 1(a). The coverage is narrow, clustered and repeatedly revisited: 46 islands concentrated near 230 and 345 GHz, with summed window bandwidth 23.9× the union. The union measures frequency coverage, the summed bandwidth integrated search effort.

quencies ALMA delivered, and every drift rate quoted is the apparent one at the telescope, since that is what the data contain and what a reader can recompute from the released channel axes. Two steps need the star’s own frame instead, and both take it: a crossing is compared with the line catalogue, and a repeat epoch is read at the predicted cell, after each block’s frequency axis has been corrected for the Earth’s motion at that epoch and then for the star’s systemic velocity. Nothing here is evaluated in a barycentric frame alone.

Two power quantities carry the main text, both defined in Table 2: the trigger level $P_{\text{trig}} = 4\pi d^2 S \Delta\nu$ at distance d , for a carrier that makes $T_\star$ reach $T_{\text{trig}} = 5$ in one native channel of width $\Delta\nu$, and the sensitivity EIRP_{90} . The second is measured, not assumed: 8860 synthetic carriers of known amplitude were injected into calibrated visibilities and recovered through the unmodified pipeline, spanning 52 block-window configurations, both ALMA arrays, four bands and the sample’s full range of channel widths. Every limit quoted in this paper is the EIRP_{90} of its own window, given to two significant figures beside its native channel width; the measured completeness is §5.1 and its error budget is Table 3. The absolute flux scale is checked twice from outside, in two bands on two arrays: the extractor reproduces an independent reduction of the same AU Mic visibilities to 0.994 ± 0.014 , and the published quiescent flux density of ϵ Eridani to 1.10 ± 0.31 (Burton et al. 2022).

The drift range searched is bounded by an acceleration, not by a frequency. Drift rate and line-of-sight acceleration are one quantity in two sets of units, $a_{\text{los}} = c \dot{\nu}/\nu$, so a ceiling expressed as $\dot{\nu}/\nu$ is the same physical limit in every band. The default ceiling is $12.0\text{--}13.3 \text{ Hz s}^{-1} \text{ GHz}^{-1}$, that is $|a_{\text{los}}| \leq 3.60\text{--}4.00 \text{ m s}^{-2}$. That is the acceleration of a transmitter orbiting a solar-type star at a few hundredths of an au. It is the Mason et al. (2025) fiducial with a threefold margin, widened for a confirmed exoplanet host whenever the innermost known planet demands more. The margin is not cosmetic: at unit margin the search would have been blind to two of the windows that produced a crossing. The grid is uniform in $\dot{\nu}$, and fine enough that a carrier midway between two trials loses no more than 0.26 of a native channel over the longest track. Two limits follow: the rates searched are the apparent ones at the telescope, so a deliberately chirped carrier is not distinguished from an accelerat-

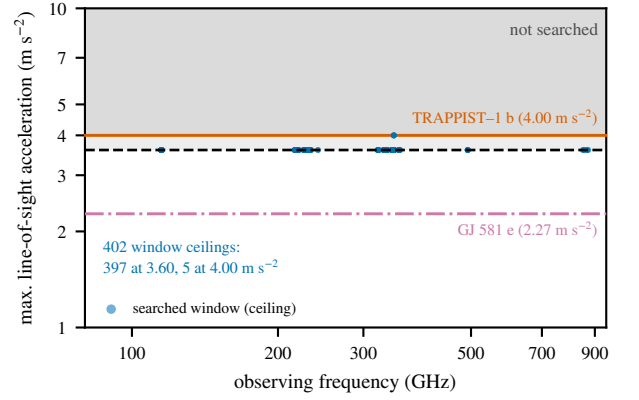


FIG. 3.— What the drift search reaches, as a physical selection function. The drift ceiling of each of the 402 Class A windows (points), converted to line-of-sight acceleration, $|a| = c \dot{\nu}/\nu$, against observing frequency; the shaded region above is unsearched. The boundary is flat because the grid was built at constant acceleration; in the units the search works in it is not, running 1.4 kHz s^{-1} at 115 GHz to 10.5 kHz s^{-1} at 872 GHz. Horizontal lines are reference accelerations: Earth’s rotation at the equator (0.034 m s^{-2}) and orbit (0.0059 m s^{-2}), and the two most demanding known planets of any target, TRAPPIST-1 b (4.00 m s^{-2}) and GJ 581 e (2.27 m s^{-2}). The boundary drawn is the default ceiling, 3.60 m s^{-2} , which is also the median; it runs to 4.00 m s^{-2} on the 5 TRAPPIST-1 windows. For the most demanding case, transiting TRAPPIST-1 b, 29 per cent of the orbital phase lies above the default ceiling and 1 per cent above the widened one.

ing platform, and a transmitter need not be bound to a known planet at all, so drifts faster than the ceiling are simply unconstrained (Fig. 3).

Only part of the archive can carry a drift search at all, and the two classes reported here are that division (Table 2). Channelisation in the sample is bimodal, with an empty gap between the modes, so the split is unambiguous, and it coincides with the physical criterion that matters: whether the maximum searched drift crosses of order one channel or more over the on-source track, $\eta_{\text{drift}} = |\dot{\nu}_{\text{max}}| \tau_{\text{track}} / \Delta\nu_{\text{ch}} \gtrsim 1$. Each window’s own η_{drift} , trial-drift count and acceleration ceiling are released catalogue columns, so a reader may reclassify on that criterion instead; the two do not partition the survey identically. **The two classes are physically different experiments**, results are given per class, and they are combined only where that is statistically appropriate.

4.2. From a crossing to a confirmed signal

Every window passes through the same three tests in the same order, and three words name what comes out of them: a *crossing*, an *unattributed crossing* and, had one survived, a *confirmed signal* (Fig. 7). Each is set out below as the quantity it measures, the number that quantity is compared with, when that number was fixed, and what passing it does and does not establish. There is no fourth test, and none of the three is a significance test.

- (i) *Trigger* $\rightarrow$ a *crossing*. The quantity is T_* , the de-drifted statistic of Eq. 1 read at the stellar position; the criterion is $T_* \geq T_{\text{trig}} = 5$ in at least one channel $\times$ drift cell; and both the statistic and the level were fixed before the archival search began. **T_{trig} is a trigger threshold and not a detection significance:** T is not shown to follow any standard distribution under the null, so the level carries no tail probability, and a window holds between 112 and 6.87×10^6 independent cells, so one level buys very different protection in a coarse single-pointing window and in a finely channelised one on a long track. A crossing establishes only that the window goes on to the next two tests.
- (ii) *Line attribution* $\rightarrow$ an *unattributed crossing*. The quantity is the crossing’s velocity offset from the nearest transition of a molecular-line catalogue, evaluated in the star’s own rest frame; the criterion is $\pm 50 \text{ km s}^{-1}$. The statistic, the trigger and the original species list were fixed before any crossing was classified; the species list was afterwards extended, and that extension is the reason one crossing is reported as unattributed where the list as it now stands would attribute it (Appendix C). The half-width follows the Keplerian line-of-sight velocities of gas surviving beyond a few au, with margin for the systemic velocity. A crossing inside that window is *attributed*, and the attribution is good evidence: a coincidence with a catalogued transition makes an astrophysical origin far more plausible than a transmitter, the more so where a second transition of the same species or known circumstellar emission is present. But a coincidence is a statement about a catalogue rather than a property of a frequency, so this is an attribution and not an exclusion, and **every crossing goes on to the recurrence test whichever side of the window it falls**. An unattributed crossing is a target for follow-up and not a detection. The window’s price is that the survey constrains nothing inside it: $\approx 113.3 \text{ GHz}$ of the 118.1 GHz covered lies outside (Appendix C).
- (iii) *Independent recurrence* $\rightarrow$ a *confirmed signal*. The quantity is T_* at the cell the first epoch predicts, read in a later execution block of the same star at a single drift rate after the frame correction of §4.1; the criterion is that it reach the trigger again; and the test was specified before the repeat epochs were searched. It is a weak criterion, and the archive’s uneven temporal sampling is why: only part of the sample has epochs separated well enough to apply it at all.

Two diagnostics hang off the second test, and the word is exact: each says which crossing deserves attention

first, and neither decides anything. The first ranks T_* against the same statistic at 512 control positions in the same window, drawn once from a fixed seed into an annulus placed by each block’s own synthesised and primary beams so that no control sits on the star; its lowest value, $1/(512 + 1)$, means the star beat every control. It is a ranking filter and not a significance test, because the stellar and control positions are not demonstrably exchangeable: out of sample, on 315 windows the frozen pipeline had never seen, the stellar rank has median 0.405 where one half is expected if the two were interchangeable, and a parameter-free correction for the annulus geometry does not restore an exchangeable statistic (Appendix B). It also responds to compactness rather than to origin, as β Pictoris shows: genuine carbon monoxide ranks below its own control ring there, because the resolved disc lifts the ring along with the star. The second diagnostic is a visibility-domain fit, which asks whether a crossing’s emission is at the stellar coordinates rather than elsewhere in the field; it cannot discriminate at the trigger either, because a noise maximum at the star has flux at the star (§5.5). **Neither gates a disposition, so neither is charged to the sensitivity**, and the limits here are correspondingly deeper than a chain charging for the rank would give. Throughout, p denotes a genuine significance level, and no probability computed against the rank’s own null is called one.

One execution block is not a null, and a rule fixed in advance says so. The comparison with the control positions needs each block’s ensemble to behave like noise, and a block fails when the median control statistic over its own windows, scaled by the median of that quantity over its class, exceeds twice that scale. Of the 404 primary-census blocks, 1 fails — A002_Xc7fa6f_X28a8 toward BD+05 1668, at 4.1 where the next worst is 1.12. **The Class A results are identical with it and without it**, the same 402 windows and 52 crossings either way, because its 4 windows are all coarse-channel; what it does hold is 4 of the 56 crossings, the survey’s entire Class B crossing population (Appendix D).

Because a fixed threshold is not a fixed false-alarm rate, the number of crossings expected by chance is computed window by window, from each window’s own cell count and the exceedance rate measured at its own control positions, and then scaled by the factor by which a star’s own tail exceeds its controls’, which the hold-out measures because the two are not exchangeable (§5.3.1).

Three corrections precede the statistic: the extraction field is chosen so that the star lies inside the delivered primary-beam field and not merely inside the quoted field of view; the stellar position is propagated to the observation epoch; and on-source time is accounted per integration from the delivered exposure records rather than from the scan intentions. That the chain finds what it is meant to find is shown twice over, by the injected carriers above and by β Pictoris CO — a real feature at a known stellar position, recovered and localised in two transitions and 9 execution blocks (§5.4).

5. RESULTS

The primary experiment is the Class A search for a drifting, unresolved spectral carrier: 402 fine-channel windows toward 60 systems. The Class B search for an unresolved spectral excess is secondary and carries

TABLE 2

THE QUANTITIES AND THE THREE WORDS THIS PAPER USES, DEFINED ONCE. EVERY TERM BELOW IS USED IN THIS SENSE THROUGHOUT AND IN NO OTHER.

Term	Definition
Crossing	A channel $\times$ drift cell at the stellar position reaching $T_\star \geq 5$. An event requiring investigation; no claimed significance.
Unattributed crossing	A crossing lying more than 50 km s^{-1} in the star's frame from every transition of the frozen line catalogue. A target for follow-up; not a detection.
Confirmed signal	An unattributed crossing recovered at the predicted stellar-frame frequency and drift in an independent block. The only confirmation criterion.
P_{trig}	The power of a carrier that makes $T_\star$ reach the trigger in one native channel. An internal pipeline level: not a sensitivity, not a significance.
EIRP ₉₀	The power at which nine injected carriers in ten are recovered at the star through the trigger. A measured completeness, not a detection probability.
Class A	The primary experiment: 402 fine-channel windows, median 488.3 kHz, toward 60 systems over 47.7 GHz of union bandwidth, in which the searched drift crosses of order a channel or more.
Class B	The secondary experiment: 1249 coarse-channel windows, mode 15.625 MHz, constraining a persistent unresolved excess. No drift discrimination, and never a carrier search.
The sample	Neither volume-limited nor representative of the solar neighbourhood: the stars ALMA happened to observe for other reasons (§3).

no drift discrimination: 1249 coarse windows. Together they are the 1651 windows of the released catalogue. **No event satisfied the confirmation criterion set out in §4.2.** What follows is first how deep the search reaches, and then what it found and how each finding was disposed of.

5.1. Sensitivity: the power at which a carrier would have been found

The sensitivity of this survey is a measured completeness and not a threshold. Artificial carriers were added to the real calibrated visibilities, each at a random sub-channel phase and with a random drift rate from its window's own grid, and recovered through the unchanged pipeline: extraction, baseline removal, de-drifted stack and the trigger at the star. The power at which nine carriers in ten come back is EIRP₉₀, the only sensitivity quoted here. *The completeness is measured through the chain the survey applies* — the trigger and localisation at the stellar position — and the 512-position spatial rank, which annotates a crossing rather than deciding it (§4.2), is not charged against it. Requiring a carrier to outrank every control position as well would cost a further factor 1.31, and in 3 windows with a bright ring no injected power reaches ninety per cent at all (Fig. 4): charging a limit for a gate that disposes of nothing would make the survey look shallower than it is.

Measured that way the completeness is one number.

Nine carriers in ten are recovered at $3.03 P_{\text{trig}}$, where P_{trig} is the window's own trigger power, and the factor is determined to ± 2 per cent. Every one of the 49 directly injected Class A windows resolves a ninety per cent point, between 2.41 and $4.80 P_{\text{trig}}$, and the value does not depend on the brightness of the window's control ring: the 5 windows whose ring carries resolved emission come back at $3.24 P_{\text{trig}}$ against 2.94 for the rest, a ratio of 1.10 . One factor therefore serves the class, and weighting the curve by the survey's own distribution of ring brightness instead of pooling moves it by 1.1 per cent, in the direction that makes the pooled factor conservative. Class B carries its own factor $4.17 P_{\text{trig}}$.

The resulting limits. The median Class A window reaches $2.9 \times 10^{15} \text{ W}$ of equivalent isotropic radiated power and the median system, on its best window, $1.6 \times 10^{15} \text{ W}$, with a spread across systems of 6.8×10^{13} – $3.7 \times 10^{16} \text{ W}$ (Fig. 5a). The round 10^{15} W level is reached for 25 of them. Table 3 collects the terms acting on those numbers and the two-sided ones, which combine to $\times 0.89$ – $\times 1.12$ on every limit.

Carriers were injected directly into 49 of the 402 Class A windows, 12 per cent, and the result transferred to the remaining 353, which is the dominant caveat on all of the above. The injected windows are the Class A windows of execution blocks drawn in a stratified random order fixed in advance of the measurement, from the whole catalogue and without reference to which windows hold a crossing; they hold 7 crossings against the 6.3 the population rate predicts. *The unit drawn is the block and not the window*, because a block's calibrated visibilities are what an injection campaign has to rebuild: band and array are properties of a block and were controlled by the draw, while channel width, control-ring brightness and integration length vary inside one and are compared with the population instead, the largest discrepancy on any axis being 22 per cent. The comparison matters most for the ring, bright in only 3 per cent of the Class A catalogue and so easy for a small sample to misrepresent. The median ninety per cent point differs by 0.18 between bands and 0.24 between channel widths; the largest difference is between the arrays, 3.10 on the compact array against 2.90 on the 12-m array. *Most of that is the control ring and not the array*: setting the bright-ring windows aside leaves 0.20 , so 0 per cent of the apparent array dependence is the ring, and the residue is consistent with a compact array whose shorter baselines keep more of each field's extended emission inside the beam. *That separation is only possible at this sample size.* In a fifth as many windows the two are confounded — the windows with a bright ring sit almost entirely on one array — and the same comparison then attributes the whole difference to the array. It is a plain illustration of why a completeness measured on a handful of windows and transferred to four hundred has to be checked by injecting into more of them.

One geometric term acts window by window, and it is measured rather than modelled. Whether neglecting the annual parallax is a simple loss of amplitude depends on the displacement in synthesised beams: a small fraction of a beam only attenuates a compact source, while a displacement approaching one puts the phase wrong on the longest baselines, where no scalar can describe it. Over the 1651 released windows the median displacement is

TABLE 3

WHAT SETS EIRP₉₀, AND IN WHICH DIRECTION. THE TWO-SIDED TERMS COMBINE IN QUADRATURE TO ONE INTERVAL APPLYING TO EVERY LIMIT; THE SIGNED TERMS ARE BIASES, STATED WITH THEIR DIRECTION RATHER THAN FOLDED INTO AN ERROR BAR. THE CHANNEL RESPONSE IS CARRIED BY THE INJECTIONS RATHER THAN APPLIED AFTERWARDS, SO IT MUST NOT BE APPLIED AGAIN (APPENDIX A).

	value	what it does to the limit
Measured completeness	$\times 3.03$	nine carriers in ten recovered at this multiple of a window's own trigger power, over 49 injected windows
Two-sided uncertainties (per cent), combined in quadrature		
Window-to-window transfer	$-6/+8$	the central two thirds of the 49 injected windows; full range $\times 0.80$ – $\times 1.58$, and the factor itself is determined to ± 2
Flux, distance and visibility scale	± 9	calibration
Combined	$-11/+12$	$\times 0.89$ – $\times 1.12$ on every limit
Signed terms. A bias is not an error bar and is not added to one.		
Channel response	$\times 1.93$	carried by the injections, so already inside the completeness above
its residual	$+8$	the injected profile falls short of the exact response: <i>conservative</i>
Primary-beam model	$+0/+10$	ALMA's beamwidth in place of the pipeline's Airy-null one: nothing in the median window, 4 windows of one star
Decorrelation	$-12/+17$	declared from the literature, not measured here

0.022 of a beam and the largest 0.67, with 68 windows above a quarter of one; extracted at the full apparent position those leave each window's own noise unchanged to 0.20 per cent and alter no disposition. Elsewhere the retained fraction of the dirty beam divides each limit, making it shallower by 1.5 per cent in the median Class A window, and that fraction is measured rather than assumed: injection and recovery at both positions from the same visibilities reproduces the beam model to 3 per cent at a quarter of a beam and exceeds it by 20 per cent at the largest displacement here (Appendix A). **The bias runs the opposite way to the one a reader would assume: treating the displacement as a scalar loss understates this survey's sensitivity.** One window, Wolf 28 in A002_X9f2ff8_X25b0, has no astrometric keys and is left uncorrected.

5.2. The nearest stars: Barnard's Star and Wolf 359

Barnard's Star is the second-nearest stellar system and Wolf 359 the third if brown dwarfs are excluded, and to our knowledge these are the first millimetre or submillimetre technosignature limits for either (Enriquez et al. 2017; Price et al. 2020; Margot et al. 2023). Both are non-detections. *Both limits come from the secondary Class B experiment and not from the carrier search:* the archive holds only coarse Band 6 windows toward these stars, so what is constrained is an unresolved spectral excess and not a drifting carrier. On the EIRP₉₀ convention of §5.1 Barnard's Star reaches 2.7×10^{14} – 4.0×10^{14} W over 12 windows and Wolf 359 3.1×10^{14} – 4.5×10^{14} W over 12.

5.3. Threshold crossings and how each was disposed of

The result is a chain of three tests, and it is read in one direction. Of the 1651 windows, 56 contain a channel $\times$ drift cell at the stellar position reaching $T \geq 5$. Such a cell is a *threshold crossing*. The molecular-line mask, ± 50 km s⁻¹ wide and evaluated in each star's own rest frame, identifies 20 of them with a catalogued transition. 36 are unattributed by that test, and every crossing is in one column or the other. Of the unattributed, 2 outrank all 512 control positions in their own windows, and neither recurs in any later epoch covering its frequency.

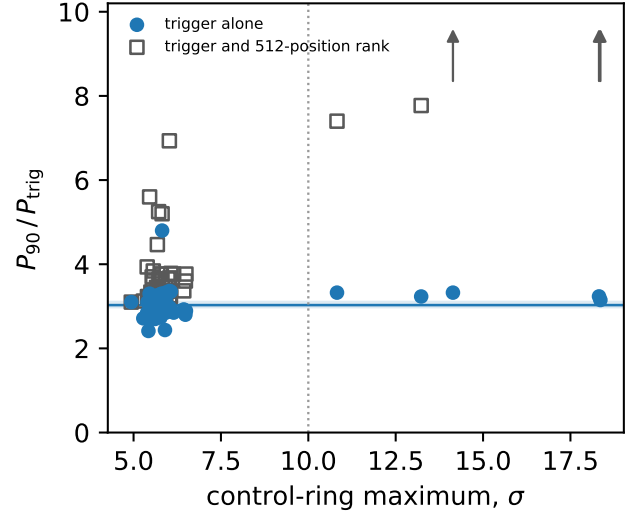


FIG. 4.— Why one completeness factor serves the whole class. Each of the 49 directly injected Class A windows is plotted at the power that recovers nine carriers in ten, against the maximum of its own 512-position control ring. Filled circles are recovery through the trigger and localisation at the star, the criterion adopted here, with the pooled value and its uncertainty as the line and band; they show no trend with the ring. Open squares add the demand that the carrier outrank every control position, and arrows mark the windows that then reach ninety per cent at no injected power. Dotted, the 10σ ring brightness above which a ring carries resolved emission rather than noise.

No crossing survives the chain, and this survey reports no detection. Figure 7 carries the chain with the number surviving each step and Table 9 the crossing-by-crossing ledger behind it.

The mask is informative rather than permissive, and its half-width is not a tuned parameter. The width follows from the Keplerian velocity bound on circumstellar gas in these systems, and between ± 20 and ± 100 km s⁻¹ the 2 crossings that outrank every control are the same ones at every width (Appendix C). The mask attributes 20 crossings where its occupancy of the searched band would give 2.3 by chance alone, so the attributions are a kinematic result and not an accident of bandwidth.

5.3.1. Most of the crossings are expected by chance

The trigger is a threshold on a statistic and not a false-alarm rate: the number of searched channel $\times$ drift cells per window spans several decades, so a common threshold buys very unequal protection. Summing each window's own measured false-alarm probability over the covered Class A windows, and correcting for the measured rate at which a star outruns its own controls, gives about 41 expected unattributed Class A crossings in a range 21–66, against 32 observed. *The number found is inside that range*, which is why the chain never treats a crossing as a detection and why no artefact hypothesis is needed to account for how many there are.

Two of this survey's own products count those crossings differently, and the difference is worth stating rather than choosing between. The released catalogue's disposition column carries 36 unattributed Class A crossings, 35 of them in windows for which an independent-cell count exists and over which the expectation above is summed; the adopted ledger has 32. Of the 35, 4 fall below the trigger once the repaired statistic is used and 2 are at-

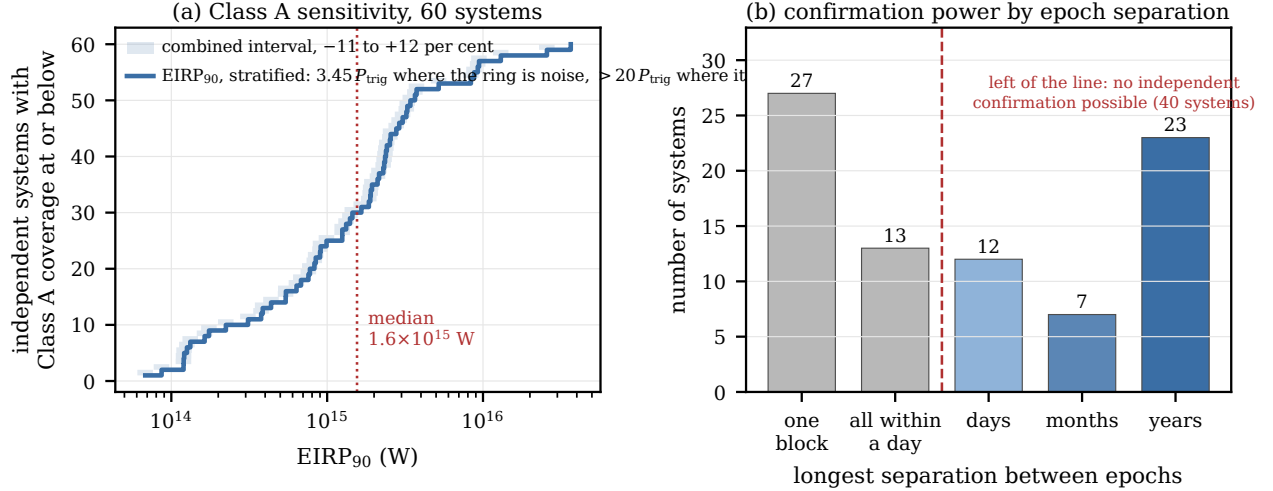


FIG. 5.— (a) Class A sensitivity: the cumulative number of systems whose best window reaches a given EIRP_{90} , the power at which nine carriers in ten are recovered through the trigger at the star. The shaded band is the combined interval of Table 3. (b) The temporal selection function: systems grouped by the longest separation between their searched epochs, an independent epoch being a separation of more than 1 day. **For the 27 systems observed in a single block, and the 13 whose blocks all fall within one day — left of the dashed line — independent confirmation was not possible at all, whatever the sensitivity in panel (a).** Recurrence over years can be tested for 23 systems of 82.

tributed by the rebuilt line list, and three crossings arrive — all of them HD 14055, one carried by the release without a cell count and 2 from the archival tail. The residue is an incompleteness of the deposited catalogue toward a single star, not a disagreement about any crossing’s disposition.

5.3.2. One execution block accounts for four of the crossings

4 crossings belong to BD+05 1668 (GJ 273, Luyten’s Star), all 4 of them in one ACA execution block, A002_Xc7fa6f_X28a8, whose whole field is above threshold: 512 of 512 control positions exceed $T = 5$ against a survey median control maximum of 2.75. The single-integration noise is as predicted (0.1609 Jy against 0.1611 Jy) but the residual does not integrate down — a jackknife on the channel mean gives 0.030 Jy against 0.0048 Jy thermal — so a slow drift common to every channel and position inflates the statistic across the field, and the 24 sibling executions of the same field, array and spectral setup reach only $T_* = 4.26$ and produce no crossing. *The median of the control ensemble, not its maximum, is the quality criterion this kind of search needs;* it is reported as a diagnostic and not adopted as a cut, because a cut strict enough to remove this block would also remove the HD 48370 carbon-monoxide control.

5.3.3. Stellar flares are excluded by a broadband prediction

Stellar activity is not excluded by the sample: its one host is classified as active, 61 Vir (rot. var.) (Astudillo-Defru et al. 2017). But a millimetre flare raises every channel of a window alike, so one able to produce a single-channel crossing at the trigger must produce a $5\sqrt{N}\sigma$ continuum event in the same integrations. Over the 48 crossings with a recoverable continuum measurement that prediction runs 49–4,717 σ , against an observed continuum statistic of median -0.29 with $|Z| < 3$ in 45 of them. One block settles it: AU Mic A002_Xa42f75_X319 holds both a crossing ($T_* = 5.27$) and the published millimetre flare of MacGregor et al. (2020), and *that real, bright, independently published flare contributes 0.17 σ*

to the single-channel statistic — raising the continuum at $Z = 10.3$ while the crossing’s own de-drifted track reaches $Z_{\text{max}} = 2.1$.

5.4. β Pictoris: an astrophysical positive control, and the obstacle it identifies

A search that finds nothing must show that it could have found something, and the survey’s strongest features show exactly that. The frozen pipeline recovers the β Pictoris circumstellar carbon monoxide blind, without being told the disc is there. It recovers it in two rotational transitions and in 10 independent execution blocks spanning 8.4 yr, and in 9 of those the star also outranks every control: 5 Band 3 windows at T_* up to 30.76, and 4 Band 6 windows at 11.68. The identification is kinematic and does not rest on those statistics. Carried through the frame chain — to the barycentre at each block’s own epoch, then to the star’s rest frame — all 10 crossings lie within 0.5 km s^{-1} of the stellar systemic velocity, in the two lowest CO transitions of a star with mapped circumstellar CO (Dent et al. 2014; Matrà et al. 2017). The Band 3 peaks agree across blocks to 1.00 channel once each block’s own tuning is removed, so any apparent drift is far below the grid searched. The alternative reading requires a transmitter placed by coincidence at matched offsets from two CO transitions of one star.

It is also the control that shows the recurrence test of §5.6 working in the positive direction: every flagged β Pictoris carbon-monoxide feature is recovered in an independent execution block, the screened Band 6 window returning $T_* = 8.95$ in the second block of its own recurrence set. A non-recurrence elsewhere means nothing unless the same machinery recovers a repeat where one exists.

One of those blocks is a counterexample, and it is the more useful half of the control. In it several of the 512 control positions exceed the stellar statistic, so the spatial rank disfavours a window holding carbon monoxide known to be real and centred on the star; the disc is resolved on the scale of the primary beam in that con-

figuration, which is the geometry the statistic is built to reject. **The rank is therefore a test of compactness and not of authenticity**, and on a real extended source it errs conservatively — a stronger statement than a clean positive control, because it bounds the diagnostic’s reach as well as demonstrating it (Fig. 11).

HD 48370 is the complementary case and is equally instructive. Its crossing reaches $T_\star = 26.83$, but the control ring rises with the star, to 18.36, as emission filling the primary beam does. The velocity matches the prominent CO(2→1) that Cataldi et al. (2023) report toward the source and flag as possible foreground cloud emission. The block’s $^{13}\text{CO}(2\rightarrow1)$ window is bright at the ring and faint at the star. Extended line emission is therefore rejected by the same spatial statistic that promotes β Pictoris.

Taken together, these two results identify the dominant obstacle for millimetre technosignature searches, and it is not sensitivity. The strongest apparent signals in this survey are circumstellar and foreground molecular emission, and 20 of the 56 crossings land on a catalogued transition. The mask that removes them costs 4.3 per cent of the unique-frequency union over 12 species, and it is nevertheless the single most productive test in the chain. The warning for future work is about line lists rather than about depth, and Appendix C puts a number on it.

5.5. The two crossings that lead their own control fields

2 of the 36 unattributed crossings reach a stellar statistic above every one of the 512 control positions in their own window: 61 Vir at 346.313811 GHz and CP–72 2713 at 344.269746 GHz, both in Band 7. They are the most interesting individual events in the search, they are drawn in Fig. 6 and tabulated in Table 4, and neither is a detection. The control rank is recorded for them because it says which crossing was examined first; it disposes of nothing (§4).

CP–72 2713’s crossing lies -12 km s^{-1} from SO(8–7₇) in the star’s own frame, inside the attribution window, and is reported unattributed anyway: sulphur monoxide entered the frozen line list after this crossing had been classified, and a species added to a mask after the event it would remove cannot be used to remove it. The coincidence is a note and not a disposition (Appendix C).

Neither is present when the same frequency is observed again. 3 other execution blocks cover 61 Vir’s cell in the star’s own frame; a carrier persisting at the discovery flux would have returned $T = 9.7$ in their combination against 2.71 found, which excludes persistence at 9.3σ . CP–72 2713 needs only one repeat: the next observation of that frequency began 2.05 hours later and is 1.1 times deeper, so a persisting carrier would have given $T = 6.2$ there against 3.60 measured.

The visibility-domain re-measurement sits beside the chain rather than inside it. 51 of the 56 crossings were re-fitted by phase rotating to the star and following the recovered drift, and 41 are consistent with a point source there; but a noise maximum at the stellar position has flux at the stellar position, so at the trigger the fit returns a real part close to the trigger itself. Injected sources in the bin these crossings occupy return $\text{Re}/\sigma = 6.58 \pm 1.97$: both events are consistent with a real source and neither excludes one. The three control sets the experiment

uses are different sizes and should not be confused: the rank screen compares the star with 512 image-plane positions, 8 of those retain a full dynamic spectrum, and 12 are reachable in the visibilities. **The discriminating evidence is the line mask and recurrence.**

5.6. Recurrence, applied to every unattributed crossing

A persistent transmitter should be there the next time the same frequency is observed. That is the confirmation criterion, applied in the star’s own frame: the expected channel follows from the Earth’s motion at the repeat epoch and the star’s systemic velocity, the discovery drift is transported to it, and the survey’s own statistic is read at that one cell. Nothing is maximised, so what comes back is an amplitude with an uncertainty. The frame is not a detail: for η Crv it moves the expected cell by -80.2 channels, so a sky-frequency test would have read a channel that holds no information about the event.

The test is run on *every* unattributed crossing that any later block covers. 35 of the 36 were tested, over 430 covering repeat windows in 137 execution blocks, 36 of them from the archival extension searched with the frozen pipeline after the statistic and the mask were fixed. *Nothing recurs.* At the predicted cell and the matched drift the largest statistic over all of them is 2.48, and 4.71 maximised over every drift trial, against a trigger of 5; every tested crossing excludes a carrier persisting at its discovery flux by at least 5.0σ , median 11.6σ . Table 9 gives the test crossing by crossing: the number of covering blocks, the interval they span, the statistic a persistent carrier would have produced, the statistic measured, and the exclusion.

One crossing could not be tested at all, and is named rather than counted: HD 14055 at 331.1559 GHz in block X1001fdc_X29ae, whose discovery dynamic spectrum was not retained, so there is no amplitude to transport into the 30 later epochs that do cover it.

5.6.1. What can be searched, and what can be confirmed

A sensitivity limit and a confirmation limit are different statements, and this experiment has two extents. What it *searched* is 60 systems and 402 Class A windows over 47.7 GHz of union bandwidth. What it could have *confirmed* is smaller: 30 of those systems were observed again at the same frequency more than a day apart, over 23.3 GHz — 49 per cent of the searched union — and 11 systems over 8.5 GHz beyond a year (Fig. 8).

Being observed twice and being observed twice at the same frequency are not the same condition, and the difference is the cost of retuning between visits: of the 82 searched systems 42 were revisited more than a day later, but only 38 at a frequency an earlier block had already covered, and beyond a year 23 against 12. Across the crossings tested the usable baselines run from 1.22 hours to 1.8 yr, and for 20 of 35 the nearest repeat falls within a day. **These tests exclude a carrier persistent over hours to days where there is repeat coverage; they bound no duty cycle measured in months or years, and for the 44 systems with no second block covering a frequency the first one covered they bound nothing at all** — a crossing toward one of those could not be retested from the archive at any strength, and the null reported here is a statement about transmitters that would have been seen again.

TABLE 4

THE TWO EVENTS OF FIG. 6, THE UNATTRIBUTED CROSSINGS WHOSE STELLAR STATISTIC EXCEEDS ALL 512 CONTROL POSITIONS OF THEIR OWN WINDOW. REPEAT EPOCHS MEASURED ARE THE BLOCKS AT WHICH THE PREDICTED STELLAR-FRAME CELL WAS READ; THE INDENTED ROWS GIVE HOW MANY OF THOSE THE RELEASED CATALOGUE CARRIES AND HOW MANY BLOCKS IT LISTS AS COVERING THE FREQUENCY AT ALL, WHICH DIFFER BECAUSE THE ARCHIVAL EXTENSION IS DEPOSITED SEPARATELY. THE TWO MEASURED VALUES ARE DIFFERENT READINGS OF ONE CELL — AT THE MATCHED DRIFT, WHICH IS ONE CELL AND ONE TRIAL AND MAY THEREFORE BE NEGATIVE, AND MAXIMISED OVER EVERY DRIFT TRIAL, WHICH IS LARGER BY CONSTRUCTION. THE EXCLUSION IS FORMED FROM THE FIRST.

	61 Vir	CP-72 2713
ALMA band	7	7
Frequency (GHz)	346.313811	344.269746
Channel width (kHz)	488.28	488.28
Drift rate (Hz s ⁻¹)	+3182	-2056
T_*	5.958	5.809
Highest control T	5.65	5.68
Flux density (mJy)	118.8 ± 19.9	11.3 ± 2.0
EIRP (W)	5.1 × 10 ¹⁴	8.9 × 10 ¹⁴
Nearest transition	SO(9 ₈ -8 ₇)	SO(8 ₈ -7 ₇)
Offset (km s ⁻¹)	-178	-12
Repeat epochs measured	3	5
in the released catalogue	3	1
catalogue blocks at ν	4	1
Observed	2017 May 20	2022 Oct 2
Nearest independent epoch	2017 May 19	2022 Oct 2
separation (h)	-1.97	+2.05
Longest baseline (d)	3	2
T expected if persistent	9.70	6.30
T measured, matched drift	+1.65	+0.96
T measured, any drift	2.71	3.60
Exclusion	9.3 σ	7.1 σ
Disposition	not confirmed	not confirmed

5.6.2. The reserved hold-out and the control beyond 40 pc

Two sets of execution blocks were searched with the frozen pipeline and then reserved for calibration (Table 7); both hold data on real targets and both are reported here as searches.

The hold-out is 77 blocks and 315 windows toward 36 stars, 214 h on source, withheld by a rule fixed before the search finished and sharing no execution block with the census. It contains 8 threshold crossings, the strongest at $T_* = 8.64$, and *not one outranks its own control positions* — the fewest controls above the star in any of them is 5 of 512. 2 are attributed in their stars' own frames and 6 are not; 7 were tested for recurrence on the same code path as the census, and exactly one is present again: the strongest, HD 285968 at +9 km s⁻¹ from CO(2-1), returning $T = 5.87$ at the predicted cell of 4 later blocks. It is a molecular line, and *that it recurs is the test working in the positive direction; every other tested crossing excludes persistence by 11 to 311 σ* (Table 5).

The control beyond 40 pc is 47 blocks and 198 windows toward 22 background sources, 88 h on source, yielding 8 crossings whose strongest is $T_* = 12.7$ and none of which outranks its own controls. A systemic velocity is known for only 0 of these sources, so the set cannot be carried into a stellar frame and supports no limit; it shows that the pipeline produces the same kind of crossing, at the same rate, on fields it was not tuned for, and nothing more. The third reserved set, used to calibrate the spatial rank, is not external at all and is described in Appendix B: the out-of-sample statement this paper makes is the hold-out's.

5.7. A stellar-frame multi-epoch stacked search

Combining the epochs of a star in that star's own rest frame deepens the limit. 58 of the surveyed stars, lying in 58 separate systems, hold more than one searched execution block. Over 315 stacked groups, 2177 epoch-windows and 1426 h, the flux limit improves by a median factor 1.52 over each star's best single epoch (2.47 at the 90th percentile, 4.45 at most). 11 further stars stack as well — 10 control stars beyond 40 pc, and 1 star covered only by archival blocks that enter no census statistic — and are kept out of those figures: over all 69 stacked stars the median gain falls to 1.46, the extra stacks being shallow two-epoch ones, and no limit quoted anywhere in this paper rests on them. Every stacked limit is measured by injection into the stack itself, on the same completeness convention as the rest of the paper, and never read off a stacked spectrum as a threshold.

A fixed stellar-frame frequency is not a fixed observed frequency, so the epochs are registered before they are added. The barycentric term alone swings by a median 8.2 km s⁻¹ between the epochs of one star and by as much as 53.9 — many channels at the 488 kHz Class A width — so epochs added as observed would dilute rather than reinforce. Each epoch is transformed to the stellar rest frame, divided by its own primary-beam response and combined by inverse variance over the channels of common stellar-frame coverage, keyed on the star's catalogued identity rather than the directory the search ran in. Spectra whose archived visibility weights are pathological are excluded first (27 of 2838, which would otherwise take almost all the weight). The transform is verified on a real line: it collapses HD 285968's molecular line from a 19.26-channel topocentric spread to 0.798 channels, an rms of 0.223 km s⁻¹, over 4 epochs spanning 44 d.

The criterion was fixed before any stellar position was examined: a stacked $Z \geq 6.0$ above every stacked control maximum, no epoch supplying more than 0.60 of the weighted sum, and at least 0.50 of the signal surviving removal of the strongest epoch. Controls are stacked identically and every clause evaluated at them, which measures the whole test's false-alarm rate at 0.21 per cent against a 1 per cent ceiling fixed in advance.

The gain is $\sqrt{N_{\text{eff}}}$ and not $\sqrt{N}$, where $N_{\text{eff}} = \sigma_{\min}^2 \sum_e \sigma_e^{-2}$ is the equivalent number of equal-depth epochs and the median N_{eff}/N is 0.65. Measured against $\sqrt{N_{\text{eff}}}$ the combination is ideal to 0.992 in the median, with a per-stack spread of ± 7 per cent that is the sampling error of a noise scale measured over 127 channels and nothing to do with stacking. Against $\sqrt{N}$ it is 0.78, the shortfall being unequal epoch depth, so an epoch count alone misleads: HD161868 stacks 13 epochs for a gain of 1.15, because its N_{eff} is 1.7.

The deepest stacked limit is Proxima Centauri's, and it is a Class B limit. Proxima holds 66 blocks — 25 primary-census blocks together with 41 archival blocks observed after the census was frozen — over 85 h spanning 6.3 yr, all of them coarse 15.625 MHz windows, so the stack constrains an unresolved spectral excess and discriminates no drift. Its N_{eff} is only 7.9, so what combining those epochs buys is $\sqrt{N_{\text{eff}}} = 2.80$. Carriers injected into every epoch and recovered through the same path give a 90 per cent recovery point at 3.84 $\times$ the stack's nominal threshold — a property of the

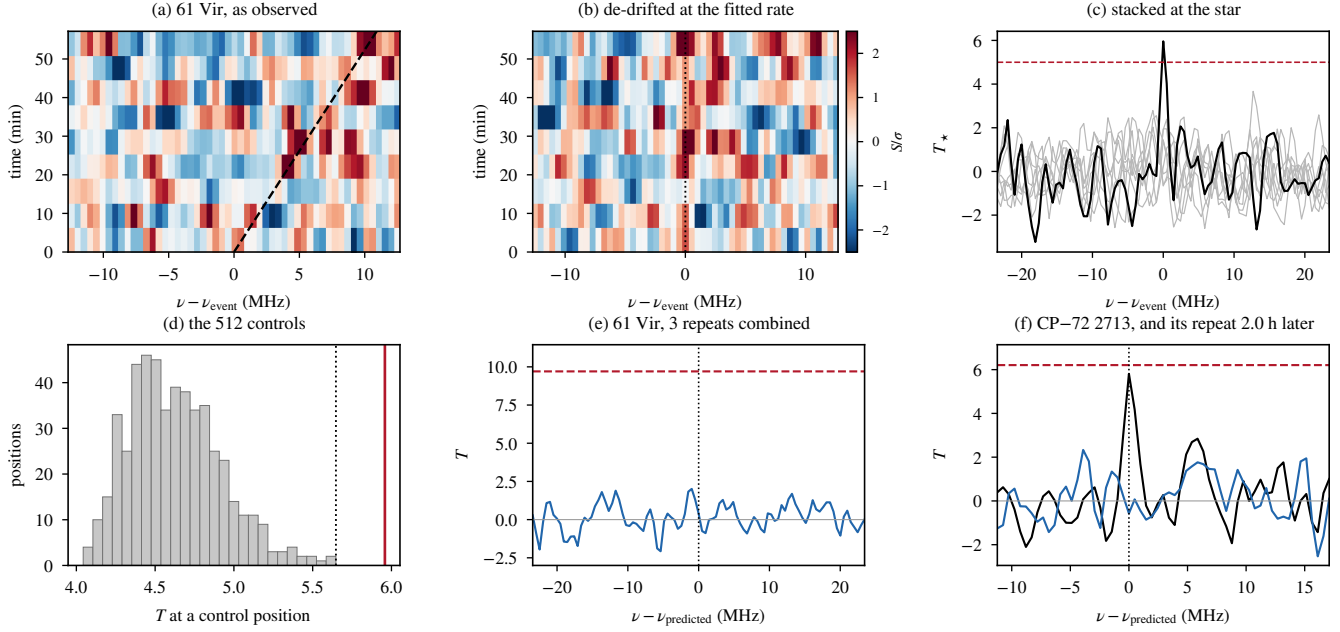


FIG. 6.— **Why this survey reports no confirmed signal although it produced an interesting event.** (a) The dynamic spectrum at 61 Vir’s position around the crossing at 346.313811 GHz, observed 2017 May 20, with the fitted drift of $+3182 \text{ Hz s}^{-1}$ as the dashed track; the 225 integrations are binned into 9 and plotted in units of each bin’s noise. At $119 \pm 20 \text{ mJy}$ against 303 mJy per integration the carrier is 0.39 standard deviations in any one of them and is invisible by eye, which is the point: the event exists only in the 38-minute coherent stack. (b) The same rows shifted by that drift, which carries the carrier 22 channels of 488.3 kHz across the observation; 8 of the 9 bins are positive at the marked channel. (c) The de-drifted stack at the star (black) and at the 8 control positions with retained dynamic spectra (grey), in units of its own noise; dashed, the trigger. (d) The same statistic at each of the 512 rank-screen control positions: the star reaches 5.96 (solid) against a highest control of 5.65 (dotted). (e) The 3 other blocks covering that cell in the star’s rest frame, combined, with zero at the *predicted* channel, which the frame transport moves between epochs; the nearest is 1.97 hours earlier than the discovery (2017 May 19), the farthest 3 days. A persisting carrier would have returned $T = 9.7$ (dashed) against 2.71 present, excluding it at 9.3σ . (f) The second event, CP-72 2713 at 344.269746 GHz: its discovery stack (black) and the same cell 2.05 hours later in a block 1.1 times deeper (blue), where a persisting carrier would have given $T = 6.2$ (dashed) against 3.60 measured. The inferred isotropic equivalent powers are $5.1 \times 10^{14} \text{ W}$ at 8.5 pc and $8.9 \times 10^{14} \text{ W}$ at 36.7 pc.

TABLE 5

THE THRESHOLD CROSSINGS OF THE RESERVED HOLD-OUT, SEARCHED WITH THE SAME FROZEN PIPELINE AND DISPOSED OF BY THE SAME TESTS AS THE CENSUS. Δv_* IS THE OFFSET FROM THE NEAREST MASKED TRANSITION IN THE STAR’S OWN FRAME AND n_{rep} THE NUMBER OF COVERING REPEAT BLOCKS; T_{pers} AND T_{rep} ARE AS IN TABLE 9. THE ONE CROSSING THAT IS PRESENT AGAIN IS A CARBON MONOXIDE LINE.

Star	block	ν_{topo} (GHz)	T_* line	Δv_*	n_{rep}	T_{pers}	T_{rep}	disposition
HD 285968	Xdaef5a Xdd9b	230.5081	8.642 CO(2–1)	+8.7	4	11.6	+5.49	attributed; present again
HD 14055	X107376c X6126	330.7123	5.513 ^{13}CO (3–2)	+112.6	30	312.0	+2.64	unattributed; does not recur
TRAPPIST-1	Xd0e450 X259b	346.2454	5.326 SO(9 ₈ –8 ₇)	-306.6	5	11.1	+0.58	unattributed; does not recur
HD 172555	X107fa96 X1300f	344.6066	5.144 SO(8 ₈ –7 ₇)	+251.3	—	—	—	unattributed; no repeat coverage
HD 285968	Xdb4b9a Xa7c	229.9129	5.106 CO(2–1)	-768.0	4	11.2	+0.22	unattributed; does not recur
HR 1010	Xc6b674 X23f2	346.2301	5.044 SO(9 ₈ –8 ₇)	-238.7	7	25.9	+0.98	unattributed; does not recur
HD 10647	Xff99e1 X9cdf	345.7059	5.017 SO(2 ₃ –2 ₁)	+37.1	13	142.1	+1.08	attributed; does not recur
HD 170773	Xff294f X1fb	345.5597	5.001 SO(2 ₃ –2 ₁)	-113.8	4	59.3	+1.78	unattributed; does not recur

stacking chain rather than of this star, since a second group returns $3.56\times$ — and on that one convention the best single epoch reaches $5.51 \times 10^{13} \text{ W}$ and the stack $\text{EIRP}_{90} = 1.73 \times 10^{13} \text{ W}$, a ratio of 3.19. The excess over $\sqrt{N_{\text{eff}}}$ is this spectrum’s own noise, 0.88 of the inverse-variance prediction across its 126 common channels, well inside the spread quoted above and not a gain beyond the root- N law. The limit holds for an emitter steady in the stellar frame to 20.9 km s^{-1} — one 15.625 MHz channel — over epochs spanning 6.3 yr, with no inter-epoch drift above $7.8 \times 10^{-2} \text{ Hz s}^{-1}$. At 1.30 pc that depth is mostly geometry; the gain is what stacking bought.

2 groups pass the full criterion and both are known lines. They are β Pictoris’ CO $J = 1-0$ at $+3.09 \text{ km s}^{-1}$ over 8 epochs ($Z = 45.3$) and its $J = 2-1$ at $+3.55 \text{ km s}^{-1}$ ($Z = 12.1$), agreeing to 0.46 km s^{-1} or 0.73 of a CO(1–0)

channel: the astrophysical positive control passing in the stack. Attributing every survivor against the survey’s own $\pm 50 \text{ km s}^{-1}$ mask leaves 0 unattributed. The stack is also an independent test of the threshold crossings of §5.6: the 9 stars carrying them stack to $Z = 2.82-4.43$ in 8 cases, 0 exceed their own control ensembles and 0 passes the criterion. The exception, HD 285968, reaches $Z = 12.89$ on the foreground line used above as the registration standard and is rejected because its own controls reach 14.19: the emission is extended, as a cloud is and a transmitter is not.

The restriction travels with every number above, and its width is the channel. At CO(2–1) one 488 kHz channel is 0.63 km s^{-1} , so the $\pm 30 \text{ km s}^{-1}$ reflex swing of a transmitter on a short-period giant planet spans 47 of them and de-registers itself completely, against only 1.5 channels of

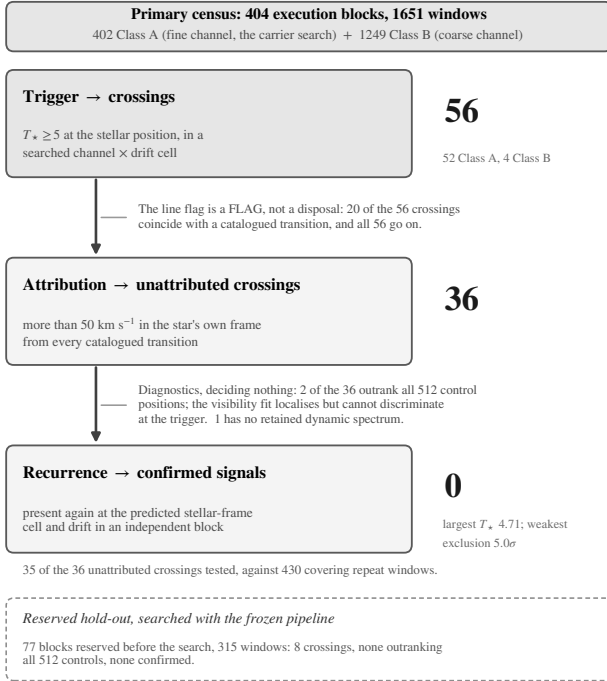


FIG. 7.— The detection logic, with the count surviving each level. Three levels and no more: a *crossing* is an event requiring investigation, an *unattributed crossing* is a target for follow-up, and a *confirmed signal* would be a detection. Coincidence with a catalogued transition is a flag and not a disposal, so all 56 crossings continue past it; the spatial rank and the visibility fit are diagnostics attached to the middle level and decide nothing. The reserved hold-out is drawn as a separate branch because a transmitter in it would matter whatever its calibration role. Every count is read from the ledger and the catalogue, so the figure, Table 9 and the text cannot disagree.

a 15.625 MHz window: the coarse stacks buy reflex tolerance with bandwidth. Drift divides the same way: in 488 kHz channels a stack is confined to $1.3 \times 10^{-1} \text{ Hz s}^{-1}$ over the median inter-epoch span of 44 d. Each stacked limit therefore constrains emitters stable in the stellar frame to within its own channel and nothing else; the single-epoch limits of §6.3 apply generally.

6. DISCUSSION

6.1. What is different about a millimetre technosignature search

The most transferable result of this experiment is not the non-detection but a measurement of what limits a carrier search at these wavelengths. For classifying a crossing in the present archival data set, molecular-line confusion and sparse repeat coverage are more immediate limitations than thermal sensitivity. That is a statement about disposing of events and not about depth: the power a transmitter must radiate to be found here remains a median $1.6 \times 10^{15} \text{ W}$ per system.

Confusion, and not interference, sets the false-positive rate. Catalogued molecular transitions, almost all of them carbon monoxide, account for 20 of the 56 threshold crossings, and no crossing in this survey is attributed to interference (Appendix E). 10 of the 12 windows that reached the spatial stage are disc or foreground carbon monoxide, arising in 2 of the 69 debris-disc systems searched. The remedy that works is a velocity-space mask evaluated in the stellar frame, and it is not free:

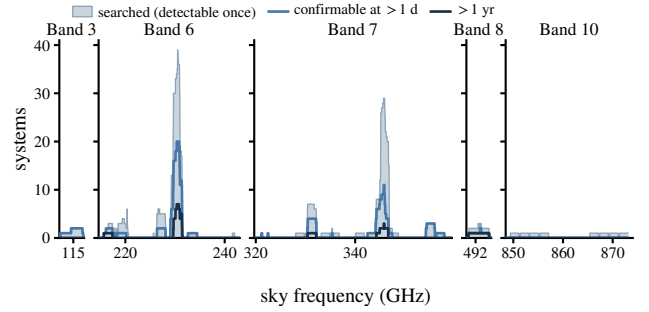


FIG. 8.— What could be searched against what could be confirmed. For the Class A experiment, against sky frequency: the number of systems toward which a carrier could have been detected once (grey), the number for which a second independent execution block more than a day later also covers that frequency (blue), and the same beyond a year (dark). One panel per populated ALMA band, widths in proportion to the frequency each contributes, as in Fig. 9. The grey curve has support 47.7 GHz of union bandwidth, the blue 23.3 GHz and the dark 8.5 GHz.

it removes 4.3 per cent of otherwise usable coverage.

Spatially extended emission is the second discriminant, and it has no analogue at centimetre wavelengths. A line filling the primary beam raises the star and the control annulus together instead of cancelling, so the spatial statistic loses power exactly where circumstellar gas is brightest. β Pictoris shows the benign face: the frozen pipeline recovers its circumstellar carbon monoxide blind, in two transitions, which is the positive control the experiment needed. HD 48370 shows the other. Its resolved disc emission lies near a band edge, and that one window contributes 59 per cent of every near-edge threshold exceedance in the survey: pooled, the near-edge rate is $\times 2.464$ the interior rate; without that window $\times 1.020$, and on a held-out half $\times 1.181$ ($p = 0.24$). What looked like a property of band edges is one disc, so no edge correction is applied to any limit.

The channelisation is inherited. 1249 of the 1651 windows have channels so coarse that every drift in the grid stays inside one channel over a whole track; they measure an amplitude but cannot distinguish a drifting carrier from a stationary one, which is why the carrier search rests on 402 of them. Native channel widths span about seven orders of magnitude across the archive (Fig. 1(b)), and nothing in the search could choose them.

Repeat epochs turn a crossing into a testable claim, and an archive supplies them only incidentally. Let the *confirmation coverage* $C_{\text{conf}}(\nu)$ be the number of systems toward which two independent execution blocks more than a day apart both cover the sky frequency ν , so that a carrier found once there could have been sought again. It must be stated against frequency rather than as a fraction of the sample, because repeat visits re-tune and a second epoch is of no use at a frequency it does not cover. Its support is 23.3 GHz of union bandwidth, 49 per cent of the 47.7 GHz searched, falling to 8.5 GHz beyond a year, and at its best near 231 GHz it reaches 20 systems (Fig. 8). The searched extent is about twice the extent over which a persistent carrier could have been confirmed, and for close to half the sample the archive constrains an intermittent emitter hardly at all, whatever its power.

Finally, a threshold is not a sensitivity. A trigger level is a property of the noise, not of what a transmitter of a given power does to a pipeline: measured by injec-

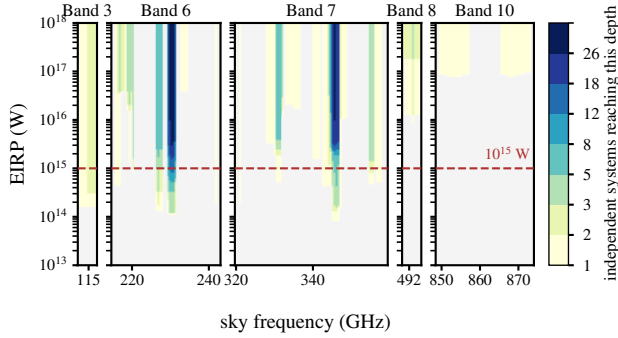


FIG. 9.— Where the experiment could have found something. Shading gives the number of independent systems that would have yielded a recovery at 90 per cent completeness for an unresolved carrier of the plotted EIRP at the plotted sky frequency; unshaded means none. One panel per populated ALMA band, with panel widths in proportion to the frequency each contributes: the 47.7 GHz Class A union is 26 disjoint islands spread across 114.0–873.1 GHz, so a single continuous abscissa would draw every island as a hairline. The dashed line is 1×10^{15} W, at which 19.1 GHz of the union survives. Figure 2 shows where ALMA looked; this shows where the experiment was sensitive.

tion and recovery through the unchanged selection, the power needed for ninety per cent recovery is 3.03 times a Class A window’s own trigger power and 4.17 times a Class B window’s.

6.2. The search space this archive opens

The carrier search is 402 fine-channel windows toward 60 stellar systems, 47.7 GHz of union bandwidth; the 1249 coarse-channel windows are a supplementary spectral-excess experiment that cannot search for drift at all. No previous search has looked for technosignatures toward stars within 40 pc at any frequency above about 30 GHz, and the one earlier millimetre search worked in Band 3 toward stars $25\text{--}277\times$ more distant (§2), so Band 3 is not searched toward nearby stars either. 20.6 GHz of the union here lies inside the 10 MHz–115 GHz interval Wright et al. (2018) adopt as the frequency axis of the search haystack.

Union bandwidth alone overstates what that buys, because different frequencies were searched toward different numbers of stars and to very different limits. Figure 9 therefore gives the quantity a reader needs instead: how many independent systems would have yielded a recovery at ninety per cent completeness, as a function of sky frequency and transmitter power. At an EIRP of 1×10^{15} W the covered frequency falls to 19.1 GHz, 40 per cent of the union, reaching 25 systems in all and 17 at the best-covered frequency near 231 GHz. With no limit on power that same frequency reaches 39 of the 60 systems, and half the sample is reached only above 3.4×10^{15} W.

Archival SETI buys sky and frequency cheaply and buys confirmation not at all. Across both classes this experiment reached 82 systems and 118.1 GHz of union bandwidth without an allocation, at a sensitivity no targeted programme at these frequencies has matched, on data selected by others and therefore with the biases set out in §3.1.

For comparison with other programmes the conventional measure is the continuous-waveform transmitter figure of merit of Enriquez et al. (2017), $\text{CWTFM} = \zeta_{\text{AO}} \text{EIRP}_{\text{min}} / (N_{\star} \nu_{\text{rel}})$ with $\nu_{\text{rel}} = \Delta\nu / \nu_{\text{mid}}$, smaller being better. For the Class A experiment, with $N_{\star} = 60$

and $\nu_{\text{rel}} = 0.097$, it is 3.2×10^5 on the sample’s largest EIRP_{min} and 1.3×10^4 on the median system, against 0.85 for Enriquez et al. (2017) and 0.11 for Margot et al. (2023). Nearly all of the difference is the product $N_{\star} \nu_{\text{rel}}$: 60 systems over a fractional bandwidth of 0.097 is a far smaller haystack than a wideband centimetre survey of thousands of stars. It compares surveys and is not an occurrence rate.

6.3. What this experiment excludes, and what it does not

Excluded. Consider a continuously transmitting, unresolved carrier that lies outside the molecular-line mask, inside the searched frequency range and inside the linear-drift grid, and that is present during one of the archival epochs. Such a carrier would have been found nine times in ten above the EIRP_{90} of $\$5.1$, a median of 1.6×10^{15} W per system on its best window. None is present toward any of the 60 systems the carrier search covers.

Four further axes bound the excluded population as sharply as power does. *Frequency*: the coverage is sparse and inherited (Fig. 9) and the median system is covered over only 1.84 GHz. *Drift and acceleration*: the grid reaches a line-of-sight acceleration of $3.60\text{--}4.00 \text{ m s}^{-2}$, which contains every known planet of the sample but one (Fig. 3); an oscillator drifting faster than that lies outside it. *Persistence*: the carrier must be radiating during one of the archival observations. *Spectral width*: the statistic is built for emission confined to one 15.3–1953 kHz channel, and anything broad enough to resemble continuum is removed with the spectral baseline.

Not excluded. A transmitter absent during the archival observation, or on for less than about a tenth of the time. At a duty cycle of one tenth, recovery falls to 71 per cent for the fine class and 83 for the coarse; those trials are undrifted, so the joint drifting-and-intermittent case is not measured at all. Emission inside the $\pm 50 \text{ km s}^{-1}$ molecular-line mask, at any strength, which costs 5.0 GHz of union bandwidth. Drift outside the searched grid, or drift curved enough to leave a channel within the track. Signals broad enough to be removed with the continuum baseline. Anything below a window’s own completeness. And any emission from the 17 566 stars within 40 pc that ALMA has not observed, which is almost all of them.

An illustrative engineering benchmark. A 12 m dish radiating 1 MW at 230 GHz with an aperture efficiency of 0.7 has a directive gain $G = \eta(\pi D \nu / c)^2 = 5.9 \times 10^8$ and so an EIRP of 5.9×10^{14} W, a depth this survey reaches for 16 of the 60 Class A systems. The same gain fixes how narrow the beam is, and detection requires the Earth to lie inside it: a solid angle $\Omega = 4\pi / G = 2.1 \times 10^{-8}$ sr, about 34 arcsec across, illuminating a fraction $\Omega / 4\pi = 1.7 \times 10^{-9}$ of the sky at a time. The example shows that EIRP is not transmitter power — the same megawatt radiated isotropically would be 10^6 W — so these limits apply to beamed transmitters whose beam contains the Solar System and say almost nothing about omnidirectional beacons.

Two blind spots. The search is Stokes I only, so the discrimination that polarisation would offer against unpolarised astrophysical emission is unavailable; ALMA’s feeds are linear, and the circularity of a circularly po-

larised carrier lies in the cross-hands, which this analysis does not extract. And the frequency coverage is the archive’s and not a choice: none of the “magic frequency” arguments that have guided centimetre-band searches falls in an ALMA band, so a transmitter deliberately placed on one lies outside this experiment by construction.

6.4. *What a dedicated ALMA technosignature experiment should do differently*

A purpose-built experiment on the same telescope would be far stronger, and the changes it needs follow from the limitations above. *Channelise finely everywhere*: the 1249 coarse-channel windows measure an amplitude but cannot distinguish a drifting carrier from a stationary one. *Choose the targets and the frequencies*: an archive weighted toward dusty discs is the worst case for molecular-line confusion, the M dwarfs that dominate the nearby habitable-zone planet census are the least covered, and windows placed deliberately away from the bright transitions would recover most of the 5.0 GHz the mask costs — all three are decisions a proposal can simply reverse. *Repeat the epochs on purpose, at the same tuning*: two or three visits separated by days and by years turn a crossing into a testable claim, and the separation is a scheduling constraint rather than extra sensitivity; Fig. 8 is what its absence costs. *Retain the visibilities*: rotating the phase centre to the star leaves a genuine point source entirely in the real part of the visibility. That asks whether the data contain a source *where the star is*, which is a stronger question than any rank against neighbouring positions in an image.

7. CONCLUSIONS

1. Three statements fix how new this range is. No published radio technosignature search above about 30 GHz has reached a star within 40 pc. The one published search in that range targeted stars at 1.01 kpc and beyond. And of the 118.1 GHz of unique sky frequency searched here in either class, 20.6 GHz lies inside the frequency axis of the Wright et al. (2018) search haystack, which defines a parameter space rather than coverage by any telescope.
2. The primary experiment is 402 fine-channel windows toward 60 stellar systems, covering 47.7 GHz of union bandwidth in disconnected intervals, searched for an unresolved carrier whose frequency drifts. A supplementary coarse-channel search for an unresolved spectral excess adds 1249 windows toward 81 systems and 90.5 GHz of union bandwidth, with no drift discrimination. Both come from one uniform reduction of 404 public execution blocks, 4.1 TB of raw data.
3. Sensitivity is a measured completeness, not a threshold. Carriers injected into the real calibrated visibilities are recovered through the unchanged selection — the trigger, and localisation at the star — and EIRP_{90} is the power at which nine in ten come back. The median system’s best window reaches 1.6×10^{15} W, with a spread across systems of 6.8×10^{13} – 3.7×10^{16} W. The rank and the visibility fit decide nothing, so neither is charged to that limit.
4. All 56 threshold crossings are reported with their dispositions. Of these, 20 coincide with a cata-

logued molecular transition, almost all of them carbon monoxide — including the β Pictoris disc, which the search recovered without being told where to look. Of the remaining 36, 35 were tested for recurrence in the star’s own frame, at every later observation that covered the event frequency: none is present again, the largest statistic at any predicted cell being 4.71 against $T_{\text{trig}} = 5$. For 31 of them that yields a quoted exclusion, the weakest 5.0σ ; for the other four no exclusion is quoted, because their discovery block fails the data-quality criterion (Appendix D.2) and a significance inherited from a defective block would mean nothing. The one crossing not tested had not retained its discovery spectrum, so there was no amplitude to carry into the epochs that do cover it. Just two of the 36 are brighter at their star than at all 512 control positions of their own field, which on its own decides nothing, and neither recurs.

5. What this survey could search and what it could confirm are different extents. The carrier search covered 60 systems, of which only 30 — carrying 49 per cent of the searched union bandwidth — were observed again at the same frequency more than a day apart, and 11 beyond a year. Across both classes, 44 of the 82 systems have no second block covering a frequency the first one covered, so a crossing toward one of them could not be retested from the archive at any strength: it would remain an untested crossing rather than a limit until a matched re-observation existed, and the null reported here does not cover that case.
6. Combining each surveyed star’s epochs in its own rest frame deepens the flux limit by a median factor 1.52. For Proxima Centauri, where $\sqrt{N_{\text{eff}}} = 2.80$ is what combining the epochs buys, the stacked limit measured by injection into the stack reaches 1.73×10^{13} W in 15.625 MHz channels — a Class B limit on an unresolved excess, carrying no drift discrimination — for an emitter steady in the stellar frame to 20.9 km s^{-1} across 6.3 yr.
7. These results constrain persistent, unresolved carriers only within the domain actually searched and only toward this sample (§6.3). The more transferable result is what such a search runs into. For classifying a crossing in the present archival data set, molecular-line confusion and sparse repeat coverage are more immediate limitations than thermal sensitivity, and a channelisation chosen by other observers for other purposes then decides what kind of search is possible at all — while the power a transmitter must radiate to be found here remains enormous.

AUTHOR CONTRIBUTIONS

G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript. R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection–recovery campaigns, and produced the released catalogue and its regeneration scripts. Both authors designed the vetting procedure, chose the masked species, carried out the statistical validation reported in §4.1 and Appendix D, and agreed the dispositions of Table 9.

FUNDING

This work received no dedicated funding. It used public archival data and no observing time was awarded for it.

COMPETING INTERESTS

The authors declare no competing interests. VBRL Holdings is a privately held company through which R.D. conducts independent research, named here solely as his affiliation; it had no role in the design of the survey, the selection of targets, the analysis, the decision to publish or the content of this paper, and provided no funding for the work.

ACKNOWLEDGEMENTS

This paper uses ALMA data from the public ALMA Science Archive. The 404 primary-census execution blocks searched span 2013 October 6 to 2025 June 11 across 65 proposal codes, which in the standard ADS/JAO.ALMA#*code* form are: 2012.1.00142.S, 2012.1.00198.S, 2012.1.00268.S, 2012.1.00385.S, 2012.1.00437.S, 2012.1.00993.S, 2013.1.00359.S, 2013.1.00588.S, 2013.1.01147.S, 2013.1.01214.S, 2015.1.00307.S, 2015.1.00633.S, 2015.1.00783.S, 2015.1.01015.S, 2015.1.01260.S, 2016.1.00104.S, 2016.1.00358.S, 2016.1.00515.S, 2016.1.00637.S, 2016.1.00803.S, 2016.1.00817.S, 2016.1.00878.S, 2016.1.00880.S, 2016.2.00015.S, 2016.2.00200.S, 2016.A.00013.S, 2017.1.00167.S, 2017.1.00200.S, 2017.1.00215.S, 2017.1.00561.S, 2017.1.00698.S, 2017.1.00704.S, 2017.1.00786.S, 2017.1.00986.S, 2017.1.01583.S, 2017.1.01644.S, 2018.1.00045.S, 2018.1.00072.S, 2018.1.00126.S, 2018.1.00461.S, 2018.1.00470.S, 2018.1.00947.S, 2018.1.01149.S, 2018.1.01545.S, 2018.1.01833.S, 2019.1.00189.S, 2019.1.01443.T, 2019.1.01517.S, 2019.1.01603.S, 2019.1.01681.S, 2019.2.00208.S, 2021.1.00485.S, 2021.1.01209.S, 2021.2.00097.S, 2022.1.00134.S, 2022.1.00338.L, 2022.1.00793.S, 2022.1.01163.S, 2023.1.00390.S, 2023.1.00487.S, 2024.1.00165.S, 2024.1.00595.S, 2024.1.00722.S, 2024.1.01095.S, 2024.A.00040.S. Per-block epochs are in the machine-readable release. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia* (Gaia Collaboration et al. 2023), processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (CDS, Strasbourg, France), of the NASA Exoplanet Archive (Akeson et al. 2013), and of the Splatalogue spectral-line database (Remijan et al. 2007).

DATA AVAILABILITY

The reference of record is a single permanent Zenodo deposit, cited by its DOI. It carries every number in this paper and the code that produced them, and contains: the 89-star target catalogue with its selection function; one row for each of the 1651 searched win-

dows, carrying the ALMA project code and execution-block identifier, frequency limits, native channel width, trial-drift grid, trigger statistic at the stellar position, control statistics of the 512-position ring and the window’s EIRP₉₀; all 56 threshold crossings with their stellar-frame frequencies, line classifications, recurrence measurements and dispositions; the search statistic and the molecular-line list as files in their own right; the injection-recovery records; the block-by-block archival verdicts and the ALMA delivery record for every block in scope that could not be searched; the diagnostic figures the printed paper refers to but does not reproduce, among them the per-system coverage waterfall and the selection funnel (`coverage_waterfall_full.pdf`, `selection_funnel.pdf`); the illustrative table of sample systems within 15 pc with a confirmed planet; and the full pipeline with the commit that produced the deposited products. Two properties of the deposit are measured rather than claimed: 139 generated products regenerate byte-identically from the frozen inputs, and 41 headline quantities are re-derived from the released catalogue alone, without the pipeline.

Three of the choices that could create or remove an unattributed crossing carry an identifier that can be checked independently of the deposit. The calibration hold-out rule of Appendix B.1 and the block list its first guard depends on were committed to the project repository as `c75069040eab52ec1336376a4dae62114bb281a3` on 2026 September 14 at 07:10 UTC, 3 days before the first reserved block was searched. The molecular-line list is a dated query of a public line catalogue (2026-10-06), and one of its species, sulphur monoxide, was not in the list on 2026 September 11 at 19:23 UTC, when the CP-72 2713 crossing was already recorded with a disposition (`f5f52e7804f1baad99515ea2d46ddd7e4e4cd546`); that ordering is why the crossing is reported as unattributed (Appendix C). The repository at <https://github.com/Tilanthi/SETI> is a working mirror and may move on.

SOFTWARE

The pipeline depends on CASA 6.7.6 (casatools/casatasks 6.7.6-14; CASA Team et al. 2022), Python 3 with astropy (Astropy Collaboration et al. 2022), numpy (Harris et al. 2020) and scipy (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through astroquery (Ginsburg et al. 2019); all are free and open-source, with exact versions recorded in the frozen release.

APPENDIX

SPECTRAL RESPONSE, FREQUENCY CONVENTIONS AND THE PRIMARY-BEAM RESPONSE

Checking a threshold in this paper needs three things: the frame every frequency is quoted in, the instrumental response that separates a channel threshold from a transmitter power, and the primary-beam factor that turns a measured noise into a flux at the star. The per-window catalogue, `catalogue/windows.csv` in the deposit, carries the columns each of them uses.

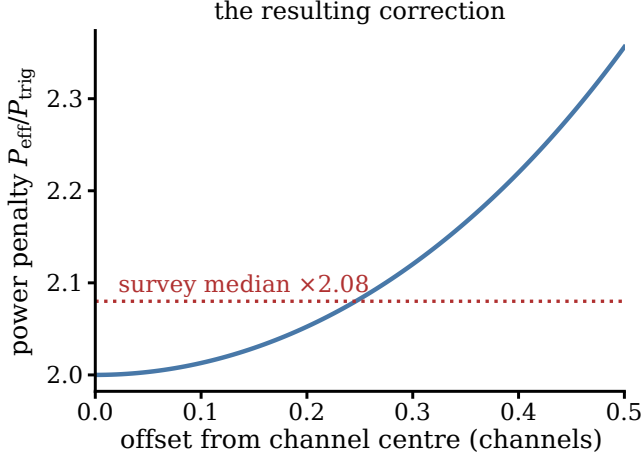


FIG. 10.— Why the nominal channel threshold is not the transmitter threshold. ALMA applies online Hanning smoothing by default, so an intrinsically unresolved carrier is spread over neighbouring channels and the peak channel holds only part of its power: half at a channel centre, 0.42 on a boundary. The threshold is set in units of the channel noise, so the power a transmitter must radiate to reach it is larger than the nominal figure by the reciprocal of that fraction, plotted here against where in the channel the carrier falls. This is the whole of the difference between P_{trig} and P_{eff} .

Frequency frames, linewidth and smearing

Every frequency axis here — searched spectra, tabulated ranges and crossing frequencies alike — is *topocentric* as delivered by the ALMA correlator, and the search itself applies no velocity-frame shift. Velocities enter at the vetting stage, where the line mask is evaluated in each star’s own rest frame: it carries every crossing frequency to the barycentre at its own block’s epoch and then to the stellar frame before comparing it with a transition (Appendix C). The residual error is then the catalogue radial-velocity uncertainty alone, a few km s^{-1} within 40 pc; second-order relativistic terms at the 50 km s^{-1} scale are $\lesssim 5 \text{ kHz}$ at 350 GHz, below the finest native channel searched. The stacking is drift-aligned and inverse-variance weighted: coherent in frequency trajectory, not in electric-field phase.

The native channel width is also the linewidth convention for every EIRP threshold. The statistic is the single-channel flux density after drift correction, so $\text{EIRP}_{\text{trig}}$ is the minimum *total* power of a channel-confined carrier, flat in the assumed linewidth below the channel and never a spectral power density; flatness is exact only for a centred tone, and a boundary-straddling one divides its power between two channels, which is the loss the injections carry and one reason EIRP values are quoted to two significant figures. The Hz-resolution surveys’ limits are total powers in the same sense, each the best case of its own channelisation, which makes the axes of Fig. 1(a) commensurable. One number fixes the scale, illustratively and not achievably: rechannelising the median Class A window’s 488 kHz to the 2.79 Hz of a dedicated narrowband search would divide the threshold by 418, and no reprocessing recovers resolution the correlator never recorded.

Channel widths come from the measurement sets’ own `CHAN_WIDTH`; MS v2 defines no “EFFECTIVE_BANDWIDTH” column, so the quoted channelisations are separations and are not claimed to be noise

bandwidths.

ALMA applies online Hanning smoothing by default (Cortes et al. 2024), so the spectral response is ~ 2 channels wide and neighbouring channels are correlated (Fig. 10). For the 0.25/0.5/0.25 kernel the coefficients are $\rho_1 = 2/3 = 0.667$ and $\rho_2 = 0.167$, with $\sigma_{\text{smoothed}}/\sigma_{\text{raw}} = 0.612$: the correlation is not confined to adjacent channels, the lag-two term being a sixth. The peak-channel fractions 0.42–0.50, median 0.48, and the resulting $\times 2.00$ – $\times 2.36$ optimism of a nominal threshold are given in §4; the released catalogue carries the nominal threshold and both corrected values for every window.

The injected carriers are deposited with the correlator’s own channel response, at a sub-channel phase drawn afresh for each tone, so the smoothing loss sits inside the measured completeness rather than being applied to it. The deposited profile is the tone divided between the two channels it straddles and then passed through the three-point kernel, which leaves 0.500 of the flux in the peak channel for a carrier at a channel centre and 0.375 for one on a boundary, against 0.4244 for the exact transform of the lag window; averaged over phase the deposition therefore falls 8 per cent short of the ideal response, and the completeness is conservative by that much. Where in a channel a carrier was placed makes no difference to its recovered signal-to-noise: the medians over 3673 tones agree across the whole range of sub-channel phase to 1 per cent, because a carrier at a typical searched drift rate crosses about 11 channels during one observation and so samples the whole response. Since the loss is carried end to end it must not be applied to the result as well: at the ninety per cent point of the recovery curve the peak channel holds $1.93\times$ the nominal trigger flux, which is that factor of about two, already spent.

Two measurements fix the smoothing state. The retained per-integration residuals of the second-stage local null have a lag-one spectral autocorrelation of 0.65 against the Hanning prediction 0.667, so the correlation is instrumental and those spectra are smoothed and un-averaged; and the archive’s reported effective spectral resolution is exactly twice the channel separation in 1505 of the 1651 windows, in both correlator modes. The other 146 report a smaller ratio, the signature of online channel averaging, where the peak-channel loss is smaller; every released window has a reported resolution, so the blanket factor is right for the great majority of both classes. The correlation model is truncated at lag two: the baseline step, a per-integration median over up to 65 channels, could imprint longer-range correlation that the released products retain no residuals to measure, which would make the independent-trials count of Appendix D an over-estimate.

Transmitters wider than one channel. The baseline filter (§4.1) is flat for features narrower than roughly half its effective width W and falls to zero above W , so the survey’s sensitivity is to channel-confined morphologies and W is the linewidth ceiling: 65–74 channels over the fine class and 30–59 over the coarse one, that is $\approx 32 \text{ MHz}$ at 488 kHz and $\approx 0.9 \text{ GHz}$ at 15.62 MHz. Within that ceiling a top-hat transmitter spanning N_{ch} native channels has per-channel flux the total divided by N_{ch} at unchanged noise, so its threshold is N_{ch} times the single-channel one, which is the rescaling the main text applies to signals broader than one channel.

Intra-integration smearing. A drift smeared by Δ channels within one integration keeps $\eta_{\text{smeared}} = \text{sinc}(\pi\Delta/2)$ of the peak amplitude, so the effective threshold is $\text{EIRP}_{\text{trig}}/\eta_{\text{smeared}}$. The excursion can be evaluated wherever the release carries the integration count, which is 451 of the 1651 windows; over those it reaches at most 1.10 channels. Only 9 of them have $\eta_{\text{smeared}} < 0.99$ — HIP 10679 Band 6 (0.574, $\times 1.74$), β Pic Band 6 (0.574, $\times 1.74$), HD 172555 Band 6 (0.574, $\times 1.74$), HIP 10679 Band 6 (0.606, $\times 1.65$), HD 172555 Band 6 (0.606, $\times 1.65$), η Crv Band 8 (0.865, $\times 1.16$), η Crv Band 7 (0.932, $\times 1.07$), HD 48370 Band 6 (0.979, $\times 1.02$) and CP-72 2713 Band 6 (0.979, $\times 1.02$) — and the other 442 agree with their nominal thresholds to better than one per cent. Compounding the worst smearing penalty with the worst Hanning factor gives $\times 4.6$ for those few, a product no released column carries.

The primary-beam response, and the two columns that let a reader recompute it

The sensitivity quoted for a window is $S_{\text{min}} = 5\sigma/A$, where A is the primary-beam response at the star's offset from the phase centre: the noise σ is measured in apparent flux, the limit is quoted in true flux, and the two differ by A . That offset is not a nuisance. Archival fields are pointed at a catalogue position, often decades old, while these are high-proper-motion stars, so the star can sit well off axis: over the 1651 released windows the offset has a median of $0.089''$, a 99th percentile of $7.43''$ and a maximum of $8.13''$, and A runs from 0.999984 in the median down to 0.5522, below 0.99 on 91 windows and below 0.9 on 34. *None* falls below one half, and every one of the low-response windows is a physically explicable J2000 pointing on a fast-moving star rather than an error. A window is kept only where the response is at least 0.5, a correction of at most $\times 2$, since at low response a small error in the beam model becomes a large error in inferred flux. The offsets are bimodal and the floor sits in the gap: the last retained window has response 0.55 at $0.46\theta_{\text{PB}}$, the first window beyond it 0.35 at $0.62\theta_{\text{PB}}$, and nothing lies between, so the rule binds prospectively and excludes none of the 1651 released windows. Beyond it lie the 12 ϵ Eri Band 6 windows, also past the first side-lobe transition where the Gaussian form fails. They are *withheld*, enter no number here and are absent from the released catalogue.

The correction is applied exactly once. The offset is computed for every measurement set from its own `FIELD::PHASE_DIR` and the propagated stellar position, so no window carries a zero-by-assumption offset, and it enters in one place only: $S_{\text{min}} = 5\sigma/A$ closes on all 1651 released rows to $8.7\text{e-}05$ in relative terms. The count is in two parts, because on 10 rows whose products are not retained A is back-solved from $5\sigma/S_{\text{min}}$ and flagged as such, so there the closure is arithmetic; the audit is a test on the other 1641 rows, where A comes from the search's own stored geometry and closes to $8.7\text{e-}05$. Nothing else needs correcting: the statistic of Eq. 1, the control ranks and the attribution are *ratios* formed inside one window, in which A divides out identically.

Three columns carry the geometry, and which is which matters. `pb_offset_arcsec` is the offset. `pb_atten` is the response applied: a Gaussian of full width $1.22\lambda/D$ — an Airy first-null coefficient — evaluated *at each win-*

dow's own dish diameter, which is reproduced over all 1651 released rows to 2.6×10^{-5} , while the same Gaussian forced to $D = 12$ m for every row departs from the column by up to 7.7×10^{-2} . The third, `theta_pb_arcsec`, is $1.22\lambda/D$ at $D = 12$ m for every window, including the 1054 taken on the $1.7\times$ smaller compact-array dishes, so A cannot be recomputed from it and `pb_atten` has to be read instead.

The beamwidth convention is a bounded systematic, and the release carries both conventions so that a reader need not take ours. ALMA's primary-beam full width is $1.13\lambda/D$, narrower than the first-null coefficient, so it attenuates more and makes every limit *shallower*: by nothing at all in the median window, by more than one per cent in 40 of 1651 and by more than ten in 4, the largest being $+10.3$ per cent — those few all LP 736-15 at $8.1''$ off axis. The lowest response in the release moves 0.5522 to 0.5005, still clear of the 0.5 retention floor but by a margin small enough to be worth saying. `pb_atten_fwhm` in the deposit carries that column window by window, with each window's own θ_{PB} beside it. Changing the convention changes no disposition, because A divides out of every ratio formed inside a window, and the published identity $S_{\text{min}} = 5\sigma/A$ is left closing on the response that formed it. The remaining model uncertainty is also bounded by measurement: CASA's blocked-Airy response is better physics than either Gaussian, the largest ratio to it over the whole release is 1.0745, and the counts of systems reaching 10^{15} W are identical under any of the three.

The star's apparent direction includes the annual parallax, and the amplitude a compact source retains when that is left out of the assumed phase centre is measured rather than modelled. The displacement decides whether the loss can be treated as a scalar at all: a small fraction of a synthesised beam attenuates a compact source and does nothing else, while a displacement approaching a beam puts the phase wrong on the longest baselines. Re-extracting the 8 windows of 2 blocks furthest out at the full apparent position — proper motion, parallax, and radial velocity where catalogued — leaves each window's own noise unchanged to 0.20 per cent and alters no disposition, but it cannot test the amplitude factor, *because the factor describes a source and those windows hold none*. The statistic at the star is a maximum over a plane of drift trajectories, so moving the extraction point by two thirds of a beam draws a nearly independent noise realisation: the ratio scatters from 0.78 to 1.15 about a median of 0.98, nowhere near the 0.49 the model predicts for a source. So a source was put there. A carrier injected at the apparent position and recovered from the *same* visibilities at both positions returns a ratio that is the normalised dirty beam at the omitted displacement, measured on that block's own baselines with no model in it (Table 6). That it exceeds the model at the larger displacement is the search's doing: maximising over a plane of trajectories lets a partly decohered carrier still find a cell, so the pipeline retains more than the beam amplitude alone and a limit corrected by the model is shallower than it needs to be.

The injection campaign is blind to A by construction, and the bias that follows is exactly zero. The injector deposits an unattenuated amplitude at the star's own direction cosines, and on all 40 scored units the ladder

TABLE 6
THE AMPLITUDE RETAINED AT A MIS-ASSUMED STELLAR POSITION, MEASURED. A CARRIER INJECTED AT THE STAR'S APPARENT POSITION AND RECOVERED FROM THE SAME VISIBILITIES TWICE: ONCE WITH THE EXTRACTION PHASE-ROTATED TO THAT POSITION, ONCE TO THE PARALLAX-FREE ONE. THE RATIO OF RECOVERED SIGNAL-TO-NOISE IS THE NORMALISED DIRTY BEAM AT THE OMITTED DISPLACEMENT; THE PREDICTION IS THE MODEL EACH WINDOW'S OWN CORRECTION USES.

star	offset (beams)	predicted	measured	tones
τ Cet	0.25	0.889	$0.920^{+0.017}_{-0.014}$	12
γ Lep	0.67	0.485	$0.583^{+0.027}_{-0.044}$	12

is referred to the *apparent* 5σ : an injected apparent amplitude recovered nine times in ten implies a true flux larger by $1/A$ while P_{trig} is the power of a true $5\sigma/A$, so A cancels identically. The cancellation is tested and not vacuous — 4 of the 40 units have apparent and corrected S_{min} differing by more than a part in a thousand. Being self-consistent in apparent flux, the campaign cannot falsify the beam model; and injected configurations reach only $A \geq 0.9102$, whereas 75 windows on 5 stars lie further off axis, down to $A = 0.552$, so a limit quoted for one of those rests on an extrapolated curve. Their 6 crossings are ratios and unaffected.

Four rows carry a catalogued noise that no retained product reproduces, and they are declared rather than reconciled: 4 windows of HD 285968 in block A002_Xd98580_X3bf6, whose catalogued σ is 0.88–1.32 per cent *higher* than the surviving product's, so the limits there are weaker and not stronger. With the 10 windows whose products are not retained at all, the declared provenance gap is 14 windows; the attenuation columns are unaffected, A coming from a retained product at the same position where $A \geq 0.999960$.

THE SPATIAL RANK, AND WHAT IT CAN SUPPORT

The spatial rank asks whether an excess sits at the star or anywhere else in the field. It is a diagnostic, no disposition rests on it, and this appendix is the measurement that settles why: The star-versus-control rank is an empirically useful way of deciding which crossing to examine first, but the stellar and control positions are not demonstrably exchangeable under the null. The rank therefore has no calibrated per-window false-alarm probability and is not used as evidence for detection.

The star lies at the phase centre, inside the control annulus, and the asymmetry is real. *It is not a primary-beam effect.* Spectra are extracted from phase-rotated visibilities in apparent flux and are never corrected for the primary-beam response (Appendix A), so the familiar rise of image noise with distance from the pointing centre cannot operate here; and were it operating, it would make the control probes noisier than the star and place the stellar rank *above* one half, the opposite of what is measured. **No mechanism for the displacement is established, and we prefer to say so.** What the control ensemble does show is a weak rise of its own level with radius, and the obvious repair is to remove it: each probe is standardised within its own radius bin, the radii being known from the fixed draw, with no free parameter.

Applied out of sample, the correction does not repair the statistic. Frozen — probe radii, bin edges and stan-

dardisation alike — and applied unmodified to 778 windows that played no part in its construction, it moves the median stellar rank from 0.409 to 0.407 against the exchangeable one half, further from it, in both out-of-sample partitions separately. Rescoring the whole survey with the gradient removed moves the list of crossings out-ranking all their controls from 12 windows to 11, losing 2 and gaining 1, and leaves the survey median rank slightly further from one half than before.

The interpretation is narrow and, we think, unavoidable. The radial gradient is real, but removing it leaves the median no closer to one half, so the residual is not radial, and neither normalisation yields a calibrated tail probability. **We therefore keep the frozen rank — it has at least the virtue of having been fixed in advance — read it only as an indication of which crossing to examine first, and rest every disposition on physical evidence or on recurrence.** The radial correction, the region-maximum audit and the pseudo-star tests each leave every disposition unchanged, and because the rank gates nothing, none of this enters a sensitivity limit.

One mechanism can be excluded by measurement, and it is the one that would matter most. Equation 1 divides every position by a single noise scale shared by the star and all its probes. A star bright enough to carry multiplicative calibration residuals of its own would then have a noise the shared scale does not describe, and its statistic would be inflated with nothing in the search to reveal it. The ratio of the two noises is measurable from the released control vectors alone: because the star and each probe are divided by the same scale and maximised over the same channel $\times$ drift grid, the stellar statistic divided by the median of the 512 control statistics *is* that ratio, and it is one when the two noises agree. Over the 1595 windows carrying no threshold crossing its median is 1.001 (star-clustered 95 per cent 0.994–1.004), and it does not climb with the brightness of the field, which is what a multiplicative error would scale with: 1.002 in the faintest fifth of control rings against 0.999 in the brightest, and 1.09 over the 8 windows whose ring exceeds 14σ and is therefore set by resolved emission rather than noise. That the ratio *can* rise is shown by the 56 windows that do carry a crossing, where it is 1.18. Part of the reason it does not rise otherwise is structural: a gain error multiplying a continuum raises every channel of an integration by the same factor, and the per-integration median along frequency removes exactly that. **A stellar-position noise denominator would therefore move no statistic in this survey.**

The reserved hold-out, and what it calibrates

The statistic and the mask were settled on the searched sample and cannot also calibrate themselves, so a hold-out was reserved by a rule fixed and recorded before any reserved block was searched: a block is held out if and only if $\text{sha256}(\text{UID})_{[8]} \bmod 5 = 0$, subject to two guards that use identifiers alone — a block on the pre-registered list deposited with this paper stays in the survey, and no system loses every block — so that the assignment can be recomputed from the UIDs and that list alone. The three choices that could manufacture a signal were fixed before the rule itself: the statistic of Eq. 1, the trigger and recurrence criteria, and then the reservation,

TABLE 7

THE BLOCK AND WINDOW LEDGER. EVERY SET OF EXECUTION BLOCKS USED ANYWHERE IN THIS PAPER, WITH THE RESULTS IT ENTERS; THE ROWS SUM, AND EACH INDENTED GROUP SUMS TO THE ROW ABOVE IT. THE PRIMARY CENSUS IS THE SURVEY: UNLESS A STATEMENT NAMES ONE OF THE OTHER SETS, IT IS COMPUTED OVER THAT ROW. HOURS ARE ON-SOURCE HOURS. CONFIRMED PLANET COUNTS, HERE AND IN THE DEPOSITED TARGET LIST, ARE TAKEN FROM THE NASA EXOPLANET ARCHIVE **PSCOMP**PARS TABLE (AKESON ET AL. 2013), ACCESSED 2026 OCTOBER 4.

Set of execution blocks	Blocks	Windows	Stars	Systems	Hours	Results it enters
Archival scope: blocks the query exposes	656	—	—	—	—	defines the scope; the sum of the next two rows
... processed with the frozen pipeline	479	—	—	—	—	the second panel below
... not processed at the freeze	177	—	—	—	—	the third panel below
Processed with the frozen pipeline ^c	484	—	—	—	—	the sum of the three rows below
Primary census	404	1651	89	82	1094	every limit, the crossing list, every chance expectation
Class A, fine channels ^a	—	402	65	60	227	the drifting-carrier experiment
Class B, coarse channels ^a	—	1249	88	81	867	the spectral-excess experiment
Reserved hold-out	77	315	36	—	214	the rank displacement, the false-alarm tail factor, the radial control profile
No window survived the quality cut	3	0	0	0	0	nothing
Not processed at the freeze: what became of each	177	—	—	—	—	the sum of the five fates below
Calibrated and searched	144	—	—	—	—	the sum of the four rows below
Archival extension , reported	115	463	16	16	269	recurrence, and the null on data the frozen pipeline had not seen
Searched after the extension was frozen	8	32	—	—	—	nothing; later than the reported extension
Withheld: primary-beam form invalid ^d	8	32	—	—	—	nothing; withheld
No window reached the export	13	0	—	—	—	nothing
No calibration in the ALMA delivery	18	—	—	—	—	nothing; the archival gap
Calibration failed	11	—	—	—	—	nothing; the archival gap
Not attempted	3	—	—	—	—	nothing; the archival gap
Working disk insufficient	1	—	—	—	—	nothing; the archival gap
Beyond 40 pc: out-of-sample control	47	198	22	—	88	the external null; no science count
External calibration sample ^b	326	1322	56	—	—	the rank calibration and the positive control; no science count

^aThe two classes partition the census windows, not its blocks: 252 blocks carry a Class A window and 389 a Class B one, and most carry both.

^bNot a subset of the archival scope: it includes blocks toward stars outside the 40 pc work list, so its blocks cannot be placed inside this ledger.

^cThis panel and the scope panel above overlap rather than add: 479 of these 484 blocks are the in-scope row above, and the other 5 carry no member observing unit in the archive metadata, so the scope query does not expose them. 3 of the 5 are in the primary census.

^dEvery one is an ϵ Eri Band 6 block, where the Gaussian primary-beam form is not valid; the same rule withholds 12 windows from the primary census.

with the first reserved block searched 3 days later. Every diagnostic built afterwards is incapable of promoting a window.

Of 484 blocks processed, 77 are reserved (15.9 per cent), giving 315 windows toward 36 stars with no star and no system lost from the headline sample. Three quantities are calibrated on those blocks and on nothing else.

First, the rank displacement. Out of sample the stellar rank has median 0.405 where one half is expected, with a block-clustered 95 per cent interval of 0.341–0.458 over 315 windows, and 14 per cent of ranks below 0.1 against 6 per cent above 0.9, so the distribution is displaced rather than growing a tail. The displacement runs the way that matters: a rank below one half places the star high, and the star in fact outranks all 512 probes in 14 of the survey’s windows against 3.2 if the positions were interchangeable. The direction therefore inflates star-first outcomes rather than suppressing them, which is why the chance expectations of Appendix D are calibrated on the measured rate. Clustering removes neither the shift nor its size: 99 of 149 blocks have a mean rank below 0.5, and a star-clustered bootstrap places the median at 0.440 (95 per cent 0.42–0.46). The searched sample leans the same way without resolving it (median 0.49, star-clustered 95 per cent 0.47–0.51), and the two distributions are indistinguishable, so the in-sample checks were underpowered rather than contradicted. A displacement that survives

a hold-out chosen before the data were seen is a property of the statistic and not of this sample, and it is the quantitative reason the rank is read as a diagnostic and never as a significance.

Second, the false-alarm tail factor — the measured rate at which a no-signal position ranks first, divided by the rate exchangeability would give. It is 1.2 on 5040 trials, with a bootstrap clustered over the 77 reserved blocks giving 0.6–1.9. That factor and its width set the accuracy of every chance expectation in this paper.

The reserved blocks were also searched in their own right, because a transmitter in them would matter whatever their calibration role: their 315 windows hold 8 threshold crossings, 0 of them outranking all 512 controls, and none satisfies the confirmation criterion.

Third, the radial behaviour of the control ensemble. The standardised control level runs +0.12, +0.10, +0.12, +0.16 from the inner to the outer edge of the annulus, a mean per-window slope of $+0.067 \pm 0.033$: the displacement reproduces out of sample, but the steep inner-edge excess seen in sample does not, so the radial term is small, of the wrong sign to account for the displacement, and removing it leaves the displacement behind.

The *external calibration sample* must not be confused with the hold-out, and it is not an out-of-sample set. Its 326 execution blocks were reserved to calibrate the rank and to supply repeats the hold-out does not have, and 263 of them are primary-census blocks and 63 hold-out

blocks, with 0 outside both: it enters no count of the science sample because its blocks are already counted there, not because it is external. Over its 1322 windows the frozen pipeline returns 7 crossings outranking all their controls — 4 of them β Pictoris CO — and 3 unattributed against 3.7 expected at the measured tail rate. The out-of-sample statement this paper makes is the hold-out’s, which shares no execution block with the census.

One crossing through the whole chain

CP–72 2713 is the clearest worked example, because every step returns a verdict rather than a shrug. In Band 7 the crossing reaches $T_\star = 5.809$ in channel 35 of its window; it is unattributed, with the one nearby transition and its provenance discussed in Appendix C; both diagnostics favour it, in that it outranks every one of the 512 control positions and the drift-following visibility fit puts the emission at the stellar position, $\text{Re}/\sigma = 5.1027$ against a control-position maximum of 1.8222; and neither of those is a promotion (§5.5).

What disposes of it is the repeat block. The second block matches the first in tuning, channel width, channel count, integration count and on-source time, registers to 0.18 of a channel, and is the deeper of the two, so a persistent emitter at the first block’s flux would have returned $T_\star = 6.21$ where it returns 3.60. Read as two measurements of one constant amplitude — the conservative reading, and the one quoted here — the pair is inconsistent at 4.6σ . The blocks are 2.05 h apart, so this is a within-night test: it excludes a persistent emitter of that strength and says nothing about an intermittent one. **The disposition rests on the recurrence test and on nothing else**, which is the chain applied to every crossing in the survey.

THE LINE MASK

A crossing that falls on a known molecular transition has an astrophysical explanation to hand, and the mask exists to say which crossings those are. It is a label and not a verdict: *every* crossing goes on to the spatial and recurrence tests whether or not it carries the label, so a transmitter that happens to lie near a transition is not excluded by construction.

The list. The mask holds every transition of the 12 species these fields are observed to emit — the three CO isotopologues, HCN, HCO^+ , CS, CN, SiO, H_2CO , SO, [C I] and hydrogen recombination — that falls inside one of the 46 searched frequency intervals with an upper-state energy below 150 K. That rule is the whole of it, and it admits 111 transitions (Table 8). No strength cut is applied: the Einstein coefficients of the CO ladder are 10^{-7} – 10^{-6} s^{-1} , so any threshold admitting the strong lines of HCN or SO would discard the brightest lines in these fields. The rarer isotopologues, ^{34}SO and H^{13}CN and the rest, are not among the named species and would add 525 transitions. The list is a Splatalogue query of CDMS and JPL (Pickett et al. 1998; Müller et al. 2005), frozen with the data release, so that the mask is reproducible as a file and not as a description.

The frame is the star’s own, and the sign is not the usual one. A crossing’s frequency is carried from the topocentric value to the barycentre using its own block’s

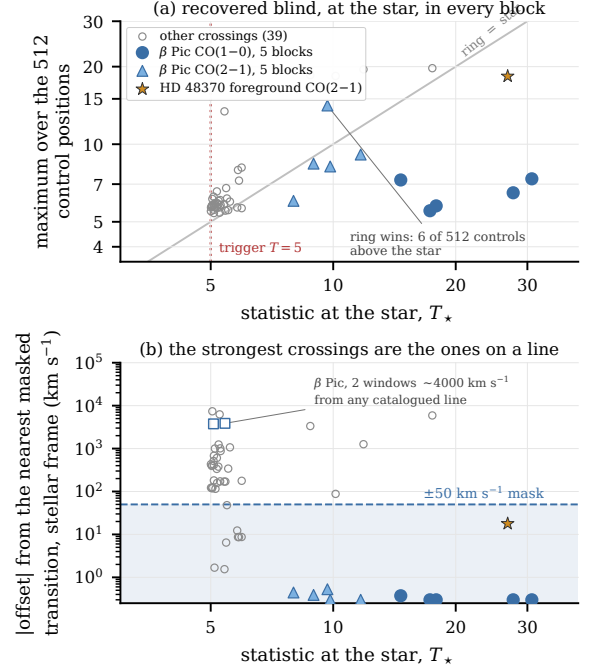


FIG. 11.— The β Pictoris positive control, in both directions. (a) The spatial screen. One point per execution block in which the blind search recovers circumstellar carbon monoxide, plotted as the stellar statistic against the maximum over that window’s 512 control positions: the 9 blocks in which the star outranks every control, and one further block in which the ring wins. *The control is that the search recovers a real narrow line at a stellar position, in two transitions and in every one of these blocks.* It is not that the star outranks the ring everywhere. In block X7116f1_X1785, Band 6, the control ring reaches 14.13 against 9.68 at the star, with 6 of the 512 control positions above it, so the screen rejects a window holding carbon monoxide that is real, centred on the star and independently mapped. The disc is resolved on the scale of the primary beam in that configuration, which is the geometry the screen is built to reject, so **the screen measures compactness and not authenticity** (§5.4). (b) Line confusion, which is what makes the screen’s job hard. Each crossing is plotted by its velocity offset from the nearest masked transition *in its own star’s rest frame*, with the adopted $\pm 50 \text{ km s}^{-1}$ half-width marked. The 10 β Pictoris carbon-monoxide crossings land 0.1 – 0.5 km s^{-1} from CO, at the resolution of the systemic velocity itself, while the two CS(5→4) crossings of the same star sit 3765 – 3870 km s^{-1} from anything catalogued and are not attributed at any width. 39 further crossings are drawn in grey, with HD 48370 marked; 52 of the 56 are plotted, the remainder being crossings for which the release stores no control ring. *A crossing is identified by where it falls in this plane and not by how strong it is*, which is why the line mask and not the trigger is the productive test.

pointing and mid-time, and then to the stellar rest frame using the star’s systemic velocity. The barycentric term runs from -29.2 to $+29.2 \text{ km s}^{-1}$ across the survey, so a tolerance applied in the observed frame would spend 58 km s^{-1} of its budget on the Earth’s motion. The 3 HD 285968 epochs show the transform working in the only way that can be checked: 12 km s^{-1} apart as observed, they agree to 0.03 km s^{-1} in the star’s frame. Throughout, $\Delta\nu_\star = c(\nu_\star - \nu_0)/\nu_0$, so a positive offset means the crossing lies *above* the transition in frequency — the opposite sign to the velocity convention of a spectral line.

The half-width, and what it costs. A one-channel test is too tight for circumstellar gas, whose emission is shifted by many channels: the tolerance follows from Keplerian motion where these molecules survive, $\lesssim 13 \text{ km s}^{-1}$ beyond 10 au, and is $\pm 50 \text{ km s}^{-1}$. Between ± 20 and

$\pm 100 \text{ km s}^{-1}$ the mask labels 18 to 21 of the 56 crossings and removes 1.9 to 7.6 per cent of the searched union, while the 2 crossings that outrank all 512 controls are the same two at every width and neither recurs. At the adopted width the cost is 5.0 GHz, or 4.3 per cent of the 118.1 GHz unique-frequency union, concentrated in the line-tuned fine-channel windows; the chain against the half-width and the cost band by band are both tabulated in the data release.

Two coincidences need saying plainly. Because the list is complete by rule it holds transitions too weak to be plausible emitters in a debris disc, and 1 crossing (HD 161868) coincides with nothing stronger; nothing turns on that, since the label does not dispose. The second is CP-72 2713, which lies -12.3 km s^{-1} from SO(8₈-7₇), well inside the mask, and is nevertheless reported as unattributed. The attribution is not astrophysically credible: sulphur monoxide has not been detected in a debris disc, where the second-generation gas is CO, C and O, and an 11.3 mJy SO line would here be unaccompanied by carbon monoxide. CO was covered. The same Band 7 window that holds the crossing holds CO(3 $\rightarrow$ 2), which falls at 345.7692 GHz in that block and is absent to 10.5 mJy at the stellar position in a 0.42 km s^{-1} channel — the same limit, in the same data, as the feature itself. The coincidence is also inadmissible procedurally, because sulphur monoxide entered the frozen list after this crossing had been examined. Reporting the crossing as unattributed keeps a crossing unattributed where the rule would have attributed it, which is the conservative direction and the one consistent with fixing the criteria in advance; the repeat block excludes a persistent carrier there in any case (Appendix B).

A wider list is not a better one. Every Splatalogue transition over the searched intervals under the same energy cut and with an Einstein coefficient above 10^{-5} s^{-1} — 5,114 transitions of 235 species — masks 75.1 per cent of the union and moves 27 of the 56 crossings across the line, one of them a crossing that outranks every control, at -36 km s^{-1} from a sulphur dioxide transition. At these frequencies most frequencies lie within $\pm 50 \text{ km s}^{-1}$ of something in such a list, which is why the species are restricted to what these fields are observed to emit, and why the outcome is a label. “Coincident with a known molecular line” is a statement about a catalogue rather than a property of a frequency.

THE FALSE-ALARM EXPECTATION AND THE CONTROL ENSEMBLE

The paper’s central claim is that the number of unattributed crossings is the number chance predicts. This appendix gives that expectation, what it is conditioned on, and what the control ensemble calibrating it can support.

What the expectation is conditioned on

The expectation is built from the control statistics themselves, which are the only empirical null the data provide, and it imposes the condition an event must actually satisfy: to reach the trigger, and to fall outside the attribution windows. Summed window by window over the 402 Class A windows, each window’s own effective cell count against its own realised null scale, 36.5 are expected to trigger by chance; the attribution windows

occupy 4 per cent of the searched band; and the product is 34.5 expected unattributed Class A crossings.

That figure assumes the star’s own tail matches its controls’, and Appendix B measures that it does not: a no-signal stellar position reaches the top of its own ensemble 1.2 times as often as exchangeability would give, with a bootstrap interval 0.6–1.9 clustered over the reserved blocks. Folding that in, the expectation is **about 41 unattributed Class A crossings, in a range 21–66, against 32 observed**. The range comes almost entirely from the width on that factor and is far wider than the difference between the two numbers, so what the comparison establishes is that the observed population needs no explanation beyond chance — not that chance predicts it to within one crossing.

The classes divide as 52 Class A crossings and 4 Class B, of which 20 and 0 respectively are attributed; so every Class B crossing is unattributed, and all of them come from the one execution block of §D.2.

One conditioning is reported only as a diagnostic. An expectation conditioned on the rank predicts a different quantity — the number of windows in which a no-signal position would rank first, 0.89 against the 2 observed — and that is a statement about the pipeline and not about the crossings. It also explains why every crossing outranking all its controls is Class A: only 455 of the 1651 windows retaining a control vector have an ensemble maximum reaching $T_{\text{trig}} = 5$ at all, and 398 of those are Class A.

One trade is not available to us, and it should be stated plainly. An excess of this size needs only a small unmodelled tail in the control ensemble, and Appendix B shows that one exists — but the same noise model sets the injection recovery fractions, so a tail large enough to remove the excess would make the quoted completeness optimistic by about the same factor. Neither correction has been applied.

Execution-block quality

The comparison with the control positions assumes each block’s ensemble behaves like noise, and one block makes that assumption plainly false: every one of its 512 probes exceeds the trigger, the smallest at 6.6. A criterion for the condition was therefore fixed and applied to every block, rather than recognised on the block that prompted it.

The statistic is the one this paper already uses as a diagnostic: the median of a window’s control ensemble, which is what separates a field with a source in it from a field that is no null at all. That median is a maximum over the window’s own channel $\times$ drift grid, so its level rises with the cell count; it is therefore scaled by the median of the same quantity over the windows of its own search class, and one threshold then means the same thing in both. A block’s statistic is the median of that ratio over its windows, and **a block fails when it exceeds twice the scale of its class** — the factor already carried by the window-level data-quality flag, fixed before that flag was applied to anything.

Applied to all 404 blocks of the primary census, where the median block sits at 1.00, the criterion fails 1: A002_Xc7fa6f_X28a8 toward BD+05 1668, at 4.1 where the next worst block is 1.12. Nothing else is close, and the unscaled form of the rule fails the same 1 block. **The**

TABLE 8

THE LINE MASK, EVERY TRANSITION IT HOLDS. n COUNTS THE HYPERFINE COMPONENTS OF ONE TRANSITION. VALUES ARE CDMS, EXCEPT THE HYDROGEN RECOMBINATION LINES, WHICH CARRY NO E_u (K).

Species	Transition	ν_0 (GHz)	E_u	n	Species	Transition	ν_0 (GHz)	E_u	n	Species	Transition	ν_0 (GHz)	E_u	n
H	41 α	92.034434	–		H ₂ CO	3 _{2,2} –2 _{2,1}	218.475632	68		SO	3 ₃ –3 ₂	339.341459	26	
SO	4 ₅ –4 ₄	100.029640	39		H ₂ CO	3 _{2,1} –2 _{2,0}	218.760066	68		CN	3 _{7/2} –2 _{5/2}	340.247770	33	5
H ₂ CO	6 _{1,5} –6 _{1,6}	101.332991	88		C ¹⁸ O	2–1	219.560354	16		SO	7 ₈ –6 ₇	340.714155	81	
H	39 α	106.737357	–		SO	6 ₅ –5 ₄	219.949442	35		CS	7–6	342.882850	66	
SO	2 ₃ –1 ₂	109.252220	21		¹³ CO	2–1	220.398701	16	4	SO	8 ₈ –7 ₇	344.310612	87	
C ¹⁸ O	1–0	109.782173	5		H ₂ CO	3 _{1,2} –2 _{1,1}	225.697775	33	7	SO	2 ₃ –2 ₁	345.704474	21	
CN	1 _{1/2} –0 _{1/2}	113.144157	5	4	CN	2 _{3/2} –1 _{3/2}	226.359871	16	7	CO	3–2	345.795990	33	
CN	1 _{3/2} –0 _{1/2}	113.490970	5	5	CN	2 _{3/2} –1 _{1/2}	226.659558	16	5	SO	9 ₈ –8 ₇	346.528481	79	
CO	1–0	115.271202	6		CN	2 _{5/2} –1 _{3/2}	226.874781	16	6	SiO	8–7	347.330581	75	
H	38 α	115.274399	–		CN	2 _{5/2} –1 _{1/2}	227.191820	16		HCN	4–3	354.505478	43	7
H ₂ CO	2 _{0,2} –1 _{0,1}	145.602949	10		CO	2–1	230.538000	17		HCO ⁺	4–3	356.734223	43	
CS	4–3	195.954211	24		H	30 α	231.900928	–		H ₂ CO	7 _{1,7} –6 _{1,6}	491.968369	106	
SO	2 ₂ –3 ₂	210.202557	19		CS	5–4	244.935557	35		[C I]	³ P ₁ – ³ P ₀	492.160651	24	
H	31 α	210.501771	–		SO	2 ₃ –3 ₂	246.404588	21		CN	6 _{11/2} –5 _{11/2}	678.813471	114	5
SO	7 ₈ –7 ₇	214.357039	81		C ¹⁸ O	3–2	329.330553	32		CN	6 _{11/2} –5 _{9/2}	680.046846	114	6
SO	5 ₅ –4 ₄	215.220653	44		SO	1 ₂ –0 ₁	329.385477	16		CN	6 _{13/2} –5 _{11/2}	680.263967	114	6
SiO	5–4	217.104919	31		¹³ CO	3–2	330.587982	32	4	SO	9 ₁₀ –8 ₈	682.684375	120	
H ₂ CO	3 _{0,3} –2 _{0,2}	218.222192	21		SO	10 ₁₁ –10 ₁₀	336.553811	143						

complete Class A results are identical with the failing block and without it — 402 windows, 52 crossings, 32 unattributed, 2 outranking all 512 controls — because its 4 windows are all coarse-channel ones. What it does carry is 4 of the survey’s 56 crossings, the entire Class B crossing population.

The control ensemble: its calibration, and its resolution

The ring geometry is set in §4.1; its calibration and its resolution are the subject here. Under that geometry the stellar ranks are consistent with the uniform distribution exchangeability implies, in both array strata — 597 12m windows and 1048 ACA windows, the latter with median rank 0.478 against the one half expected — over the 1645 of 1651 windows that retain a control vector. The ACA windows are measured on the geometry of each observation’s own antenna table and baselines rather than on a 12 m beam. Two cautions keep this from being a proof of exchangeability: a shift of less than about 0.042 in the rank distribution could not have been seen at all in samples of this size, so the strata are consistent with uniform rather than demonstrably uniform; and the diagnostic most sensitive to contamination of the inner controls, the excess of the innermost control bin over the outer annulus, is +0.038 with an interval -0.214 to +0.282 that spans zero.

The rank is not a significance test, and the reason is structural. Ranking the star against 512 control positions drawn from the same field is a good way to decide which windows deserve examination, but $1/(512 + 1) = 1.95 \times 10^{-3}$ is the *resolution* of the ensemble — the smallest rank it can express — and not the probability that noise produced the window. Against a survey-wide Bonferroni scale

$$\alpha_{\text{survey}} = \frac{0.05}{1651} = 3.0 \times 10^{-5}, \quad \frac{1.95 \times 10^{-3}}{\alpha_{\text{survey}}} = 64, \quad (\text{D1})$$

that floor is 64 times too coarse, and correlation widens it: neighbouring controls are correlated below the synthesised-beam scale, so a window’s annulus carries only $N_{\text{eff}} \simeq 30$ –43 independent spatial trials and the ensemble resolves no better than 0.023–0.032. Drawing more controls does not rescue it, the primary beam bounding how many independent positions a single

pointing contains. **The floor of Eq. D1 is the design’s central limitation: no single window in this survey can be significant on the rank alone, which is why confirmation here requires independent recurrence.**

The trials budget

A window holds 177–6 869 740 channel×drift cells (median 468), the survey 1.32×10^8 in total, and they are not independent trials: channel smoothing and a drift grid oversampled two to one together over-count them by 5.3. The control ensembles calibrate that budget without a Gaussian assumption. The same arithmetic predicts 476 of the 1651 windows carrying a control above T_{trig} and 455 do, agreeing to 4 per cent; corrected for the over-count, the observed control maxima sit +6 and +8 per cent above prediction in the fine and coarse classes independently. Departures of that size are why the operative statement is the empirical rate and not a Gaussian budget.

The residual on-star excess is not zero, and it is reported as an open systematic rather than explained away. Correlation cannot move the expected number above the trigger, only its clumping, so the excess over an independent-Gaussian budget divides in two: the cells within one crossing window, a factor 1.4 that the frequency-axis over-count predicts because one real excursion is counted about three times, and the number of crossing windows, a factor 1.3 with a one-sided Poisson probability of 0.04. The residual runs one way and carries the same sign at the controls, so a residual excess of chance crossings at the tens-of-per-cent level cannot be excluded.

One stratum departs from uniformity and it is the one that matters. The fine-channel windows’ ranks are not uniform over $n = 402$ windows: 12 of them contain an on-star peak above their own control maximum against 0.78 expected, while the coarse stratum, three times larger and predicting 2.43 such windows, contains 0. A coarse channel dilutes a narrow circumstellar line, so the class that can resolve one is the class that finds it — and by the same reasoning a genuine unresolved carrier would appear in Class A first. The departure survives removal of every fine window carrying a stellar crossing, so fine-window rank probabilities are optimistic for the class as

a whole and not only where a crossing sits.

What is left is bounded, and the bounds are null. Treating each control as a pseudo-star and ranking it against the other 512-1 gives 220 672 ranks that are uniform to the resolution the ensemble can express, and the real stellar ranks agree with them at $p = 0.38$. Any systematic radial suppression of the controls is below 10 per cent in the mean, and the star’s own residual variance equals the controls’ to better than a per cent in the rank-passing windows. Scrambling each window so that nothing coherent survives along a drift track carries 4 windows past the rank floor and changes no disposition; it fails only on HD 48370 Band 6, whose extended emission and imaging residuals are coherent across integrations and are destroyed by the scramble itself, so no scrambled-noise probability is quoted there.

RADIO-FREQUENCY INTERFERENCE SCREENING

Interference is the first alternative explanation for an unattributed crossing, so the screening this survey has, and the coverage it does not have, are both set out here.

Millimetre interferometry is an unusually poor environment for terrestrial interference, though not an empty one: ITU Radio Regulations Article 5 (International Telecommunication Union 2024) carries active allocations throughout 90–275 GHz, and WRC-19 identified parts of 275–450 GHz for land-mobile and fixed use. The specific hazard is the 94.0–94.1 GHz Earth exploration-satellite allocation used by spaceborne cloud radar — CloudSat and, since 2024, EarthCARE (Stephens et al. 2002; Wehr et al. 2023) — a documented contaminant of ALMA Band 3; 1 of the 1651 searched windows overlaps it, carrying 0 threshold crossings and 0 rank-passing events, so no result here rests on it. What favours this band is that terrestrial transmitters are sparse compared with centimetre wavelengths, that the array sits at 5000 m on a site chosen partly for radio quietness, and, decisively, that an interferometer is not a total-power detector: a signal not in the far field at the phase centre does not fringe-track but acquires a rapidly varying geometric phase across the array, and is suppressed in the correlation rather than imaged.

What screening exists, stated with its coverage, because the coverage is uneven. The allocation test above and the sky- and intermediate-frequency tests below are computed for every searched window from the released catalogue and are reproducible from it, as is the 512-position rank screen, which asks whether an excess is at the star or everywhere in the field and so serves as an interference test as well as a prioritisation. Recurrence covers only the crossings that have a covering repeat block, and the visibility fit — which asks whether an excess has the phase signature of a source at the phase centre, as a near-field emitter would not — is a consistency check rather than a filter. The correlator flagging is the observatory’s own, replayed verbatim from each block’s delivered calibration script, but it is not recorded per window in the release. *No per-channel interference flag of our own exists, and none can be recovered.*

Four measurements support the reasoning, and each is a test the population could have failed.

First, fixed sky frequency. Interference is a property of the observatory and not of the star, so a terrestrial

carrier should appear at the same sky frequency in every window that covers it. Here the one unattributed rank-passing event has a released crossing frequency; 40 further windows, toward 24 other stars in 32 other blocks, cover that frequency, and none of them carries a crossing in the same channel.

Second, clustering in sky frequency. If interference dominated the crossing population, the crossings would pile up at particular sky frequencies. They do cluster — but on the survey’s own molecular lines. The release gives a peak frequency for 51 of the 56 crossings; the other 5 are β Pic carbon-monoxide crossings, for which it gives the velocity offset from the line instead, which is why Table 9 lists all 56 while this test runs over 51. Of those, 11 fall inside the mask’s own $\pm 50 \text{ km s}^{-1}$ velocity window against 4.7 expected if each window’s peak fell anywhere on that window’s own channel grid, $p = 0.0007$. The densest band, 230.423–230.723 GHz, holds 12 crossings, of which 9 are within that velocity window of CO(2→1); and once the line-attributed crossings are set aside, the 40 that remain show no clustering at all ($p = 0.111$). *The concentration is the attribution, not a residue left over after it.*

Third, fixed intermediate frequency. An artefact of the signal chain would sit at a fixed position within the spectral window rather than at a fixed sky frequency. This test is run over every window for which the catalogue releases a peak frequency — 1397 of the 1651, of which only 51 carry a threshold crossing — because such an artefact would occupy the same intermediate frequency in all of them and not only where the trigger fires. It does not: the peaks’ fractional positions within their own windows have median 0.50 and are uniform.

Fourth, a cross-target occupancy screen over every searched window and not only the crossings. A terrestrial carrier should recur at the same sky frequency toward *unrelated* stars. Over all 3027 searched windows toward 133 stars, with each window’s peak randomised on its own channel grid, the number of cross-star coincidences within 5 MHz is 569 against 554 ± 13 expected ($p = 0.15$); among the 109 windows whose peak exceeds the trigger it is 4 against 5.2 ± 1.9 , i.e. *below* the null, and with the molecular-line mask applied 1 against 2.1 ± 1.3 . There is no excess to explain. The screen is keyed on the star rather than on the product directory, which matters: a star’s own repeat observations are not different targets, and every persistent feature — a real line most of all — would otherwise enter as a cross-target coincidence. 35 of those 109 windows sit on a masked transition (CO(2–1) $\times 17$, CO(1–0) $\times 10$, CO(3–2) $\times 4$, $^{13}\text{CO}(2–1)$ $\times 3$, SiO(5–4) $\times 1$), which is the point: a CO line falls at almost the same sky frequency toward every nearby star, so cross-star frequency coincidence is exactly what real line emission produces, and the screen cannot distinguish it from interference unless the mask is applied first.

One execution block fails these tests, and it fails them as a block rather than as a carrier. Its 4 crossings toward BD+05 1668 are the highest statistics in the survey outside β Pictoris and HD 48370, and the reason is in the field and not at the star: all 512 control positions exceed the trigger, the ring’s own median running 10.7–11.7 against a survey median of 2.74, so the window contains no null against which anything could be localised. The visibility fit returns a large imaginary part, placing the

TABLE 9

THE CROSSING LEDGER, FROM WHICH EVERY DISPOSITION IN THIS PAPER CAN BE REPRODUCED. EVERY ONE OF THE 56 THRESHOLD CROSSINGS: THE STAR, ITS EXECUTION BLOCK, THE ALMA BAND, THE CROSSING FREQUENCY AS THE CORRELATOR DELIVERED IT, THE SAME FREQUENCY IN THE STAR'S REST FRAME, THE SEARCH STATISTIC T_* , THE LARGEST STATISTIC T_{ctrl} AMONG THE WINDOW'S 512 CONTROL POSITIONS, THE NEAREST MASKED TRANSITION AND THE STELLAR-FRAME OFFSET $\Delta\nu_*$ FROM IT, AND THEN THE RECURRENCE TEST: n_{rep} COVERING REPEAT BLOCKS SPANNING Δt DAYS, THE STATISTIC T_{pers} A CARRIER PERSISTING AT THE DISCOVERY FLUX WOULD HAVE PRODUCED AT THE PREDICTED STELLAR-FRAME CELL, THE STATISTIC T_{rep} MEASURED THERE AT THE MATCHED DRIFT, AND THE EXCLUSION. T_{ctrl} IS A DIAGNOSTIC AND GATES NOTHING: IT SAYS WHICH CROSSING WAS EXAMINED FIRST. THE DRIFT-MAXIMISED READING OF THE SAME CELL IS IN TABLE 4 FOR THE TWO LEADING EVENTS AND IN THE DATA RELEASE FOR ALL OF THEM. ONE EXECUTION BLOCK FAILS THE QUALITY CRITERION OF §5.3.2, AND THE TWO QUANTITIES THAT WOULD INHERIT ITS DISCOVERY AMPLITUDE — T_{pers} AND THE EXCLUSION — ARE WITHHELD ON ITS FOUR ROWS RATHER THAN PRINTED FROM DATA THE PAPER CALLS DEFECTIVE. A CROSSING IS ATTRIBUTED WHERE $|\Delta\nu_*| \leq 50 \text{ km s}^{-1}$; THE TRANSITION NAMED IS THE NEAREST IN THAT FRAME, SO FAR FROM EVERY TRANSITION IT IS THE MAGNITUDE AND NOT THE NAME THAT CARRIES INFORMATION. ν_* FOLLOWS FROM ν_{topo} BY THE BARYCENTRIC TERM OF THAT BLOCK'S EPOCH AND THE STAR'S SYSTEMIC VELOCITY AND NOTHING ELSE, SO THE LAST COLUMN IS REPRODUCIBLE FROM THE TWO BEFORE IT. BLOCK IDENTIFIERS OMIT THE CONSTANT A002_ PREFIX.

	block	$B\nu_{\text{topo}}$	ν_*	T_*	T_{ctrl}	line	$\Delta\nu_*$	n_{rep}	Δt (d)	T_{pers}	T_{rep}	σ_{exc}	disposition
BD+05 1668	Xc7fa6f_X28a8	6	240.1016	240.105473	398.19	CS(5–4)	-5911.9	29	0.9–113	n/a+1.81	n/a	unattributed; flagged	
BD+05 1668	Xc7fa6f_X28a8	6	224.7422	224.745848	303.19	H ₂ CO(3 _{1,2} –2 _{1,1})	-1264.5	29	0.9–113	n/a+1.62	n/a	unattributed; flagged	
BD+05 1668	Xc7fa6f_X28a8	6	226.7422	226.745841	355.18	CN(2 _{3/2} –1 _{1/2})	+88.0	29	0.9–113	n/a+2.37	n/a	unattributed; flagged	
BD+05 1668	Xc7fa6f_X28a8	6	242.1953	242.199239	586.18	CS(5–4)	-3349.2	29	0.9–113	n/a+1.42	n/a	unattributed; flagged	
β Pic	Xf5d76d_Xeb8	3	115.2603	115.271130	765.7	3.34 CO(1–0)	-0.2	—	—	—	—	—	attributed
β Pic	Xf5d76d_Xe19	3	115.2603	115.271127	684.6	4.48 CO(1–0)	-0.2	—	—	—	—	—	attributed
HD 48370	Xc26103_X155a	6	230.5117	230.524326	834.18	3.6 CO(2–1)	-17.8	—	—	n/a	—	—	n/a attributed; flagged
β Pic	Xf4df6f_Xc7	3	115.2620	115.271117	907.5	5.76 CO(1–0)	-0.2	—	—	—	—	—	attributed
β Pic	Xf5d76d_Xcc5	3	115.2604	115.271117	288.5	5.52 CO(1–0)	-0.2	—	—	—	—	—	attributed
β Pic	Xf5d76d_X32f1	3	115.2606	115.271314	654.7	7.27 CO(1–0)	+0.4	—	—	—	—	—	attributed
β Pic	Xd9668b_X3a90	6	230.5161	230.537811	681.9	9.12 CO(2–1)	-0.3	—	—	—	—	—	attributed
β Pic	X6f1341_X148a	6	230.5284	230.5379	9.832	8.20 CO(2–1)	-0.1	—	—	—	—	—	attributed
β Pic	X7116f1_X1785	6	230.5272	230.5384	9.677	14.13 CO(2–1)	+0.5	—	—	—	—	—	attributed
β Pic	Xd9668b_Xa9df	6	230.5160	230.5377	8.948	8.41 CO(2–1)	-0.4	—	—	—	—	—	attributed
β Pic	Xa7a216_X23e4	6	230.5273	230.5377	7.988	6.03 CO(2–1)	-0.4	—	—	—	—	—	attributed
61 Vir	Xc079b5_X82f	7	346.3138	346.3232	5.958	5.65 SO(9 ₈ –8 ₇)	-177.6	3	0.1–3.0	9.7+1.65	9.3	unattributed; no recurrence	
HD 285968	Xd98580_X3bf6	6	230.5021	230.5447	5.956	8.12 CO(2–1)	+8.7	—	—	—	—	—	attributed
HD 285968	Xdb7ab7_X5b4e	6	230.5116	230.5447	5.868	7.22 CO(2–1)	+8.7	—	—	—	—	—	attributed
HD 285968	Xdb4b9a_X108f	6	230.5100	230.5447	5.835	7.94 CO(2–1)	+8.7	—	—	—	—	—	attributed
CP–72 2713	Xf0235_X4a6d	7	344.2697	344.2965	5.809	5.68 SO(8 ₈ –7 ₇)	-12.3	5	0.1–2.1	6.3+0.96	7.1	unattributed; no recurrence	
HD 139084 B	Xcd8029_Xb6b0	7	345.8241	345.8241	5.763	— CO(3–2)	+24.4	—	—	—	—	—	attributed
HD 14055	Xff3d8_Xd46	7	331.7699	331.7664	5.575	5.99 ¹³ CO(3–2)	+1068.7	30	0.9–650	170.4+2.35	170.4	unattributed; no recurrence	
HD 23484	X10b22d7_X2445	6	230.7904	230.8006	5.520	5.55 CO(2–1)	+341.5	4	0.1–2.1	10.4+1.53	9.2	unattributed; no recurrence	
HD 161868	Xf45a6_X10ca8	7	345.6338	345.6491	5.481	6.16 SO(2 ₃ –2 ₁)	-48.0	13	0.0–6.1	92.2+1.56	92.0	attributed	
HIP 10679	Xa95c04_X14cd	6	220.4185	220.4035	5.456	6.76 ¹³ CO(2–1)	+6.5	—	—	—	—	—	attributed
HD 14055	X11adad7_X25c5e7	7	345.4564	345.4303	5.448	— SO(2 ₃ –2 ₁)	-237.8	30	0.1–653	14.0+2.48	14.0	unattributed; no recurrence	
HD 207129	Xc19d6f_X5706	6	230.4226	230.4070	5.433	5.52 CO(2–1)	-170.3	4	0.8–425	69.7+0.86	70.7	unattributed; no recurrence	
β Pic	Xd9668b_X3a90	6	241.7512	241.7739	5.420	5.81 CS(5–4)	-3869.7	1	1.0	5.7+0.72	5.0	unattributed; no recurrence	
HR 1010	Xc6fa08_X1053	6	230.5133	230.5287	5.410	— CO(2–1)	-12.0	—	—	—	—	—	attributed
ALMA J153702653–33192492	Xf0235_X8392	6	230.5222	230.5392	5.405	13.41 CO(2–1)	+1.6	—	—	—	—	—	attributed
HD 110411	Xe45e29_Xb7d4	6	219.4538	219.4357	5.393	5.85 ¹⁸ O(2–1)	-170.2	1	10	6.7+0.62	6.1	unattributed; no recurrence	
HD 31392	X129754b_X6946	6	231.1998	231.2231	5.289	5.81 H(30 α)	-876.3	8	0.1–41	15.7+1.74	16.0	unattributed; no recurrence	
HD 14055	Xf99e1_X24be0	7	331.7078	331.7035	5.277	5.69 ¹³ CO(3–2)	+1011.6	30	0.9–652	252.2+2.14	252.0	unattributed; no recurrence	
AU Mic	Xa42f75_X319	6	230.6864	230.6705	5.274	6.07 CO(2–1)	+172.3	2310–455	6.4–0.09	6.5	unattributed; no recurrence		
HD 69830	X12209f7_X16edb7	7	322.4325	322.4444	5.259	5.42 ¹⁸ O(3–2)	-6268.5	5	0.1–29	8.9+1.69	9.4	unattributed; no recurrence	
ALMA J153702653–33192492	Xfe62c1_X398	6	216.1844	216.2036	5.228	6.02 SiO(5–4)	-1244.6	7	18–126	11.0+1.75	10.0	unattributed; no recurrence	
AU Mic	X899169_X1838	6	230.8252	230.8294	5.221	5.85 CO(2–1)	+379.0	2145–310	10.0+0.69	9.8	unattributed; no recurrence		
HD 14055	Xf99e1_X2c3bb	7	345.5256	345.5216	5.186	5.66 SO(2 ₃ –2 ₁)	-158.6	30	0.1–651	172.8+2.08	170.8	unattributed; no recurrence	
HD 14055	Xffde1_X3835	7	345.0099	345.0070	5.177	5.66 SO(2 ₃ –2 ₁)	-604.8	30	0.1–649	187.2+1.38	189.0	unattributed; no recurrence	
HD 14055	Xf99e1_X2b2ee	7	344.7007	344.6965	5.175	5.92 SO(8 ₈ –7 ₇)	+336.0	30	0.1–651	210.7+2.47	210.7	unattributed; no recurrence	
LHS 1140	Xd9d7c7_X6fd2	6	230.6355	230.6268	5.136	5.68 CO(2–1)	+115.5	4	5.0–27	7.3+1.03	6.9	unattributed; no recurrence	
HD 10647	Xff294f_Xa7f0	7	331.6570	331.6943	5.120	5.91 ¹³ CO(3–2)	+1003.3	10	0.1–592	125.8+1.33	125.4	unattributed; no recurrence	
CD–57 1054	Xcf525f_X5aa4	7	346.5099	346.5265	5.108	5.71 SO(9 ₈ –8 ₇)	-1.7	1	39	6.0–0.27	6.3	attributed	
ALMA J153702653–33192492	Xf8f6a9_X10359	6	217.7292	217.7269	5.091	5.41 H ₂ CO(3 _{0,3} –2 _{0,2})	-680.5	7	3.8–242	11.6+1.43	11.5	unattributed; no recurrence	
GJ 849	Xd9668b_X841a	6	230.8769	230.8581	5.090	5.83 CO(2–1)	+416.3	5	0.1–51	9.3–0.02	11.0	unattributed; no recurrence	
ALMA J153702653–33192492	Xfe62c1_X398	6	218.8729	218.8923	5.090	5.56 H ₂ CO(3 _{2,1} –2 _{2,0})	+181.2	7	18–126	10.5+1.52	11.5	unattributed; no recurrence	
HD 14055	X1001fdc_X29ae	7	331.1559	331.1546	5.082	6.26 ¹³ CO(3–2)	+513.8	—	—	—	—	—	unattributed; no spectrum
β Pic	Xd9668b_Xa9df	6	241.8367	241.8594	5.079	5.99 CS(5–4)	-3765.1	1	1.0	4.8–0.63	5.4	unattributed; no recurrence	
ALMA J153702653–33192492	Xf8f6a9_X10359	6	218.6711	218.6688	5.048	5.77 H ₂ CO(3 _{2,1} –2 _{2,0})	-125.1	7	3.8–242	11.6+1.18	11.6	unattributed; no recurrence	
HD 69830	X12277ed_X178c6	7	321.2005	321.2149	5.046	6.13 ¹⁸ O(3–2)	-7387.7	5	6.0–36	10.2+1.26	10.6	unattributed; no recurrence	
HR 1010	Xc66ce1_Xaa9	6	230.8348	230.8491	5.044	5.72 CO(2–1)	+404.6	7	1.1–14	10.3+0.54	11.8	unattributed; no recurrence	
61 Vir	Xc067f7_X822	7	345.5599	345.5678	5.038	5.79 SO(2 ₃ –2 ₁)	-118.5	3	2.0–3.0	9.3+0.60	8.8	unattributed; no recurrence	
HD 170773	Xf0235_X8c42	7	331.0101	331.0234	5.029	5.75 ¹³ CO(3–2)	+394.8	4	0.9–113	74.8+2.25	73.3	unattributed; no recurrence	
emission away from the disk centre	Xf8f6a9_X10359	6	217.7292	217.7269	5.091	5.41 H ₂ CO(3 _{0,3} –2 _{0,2})	-680.5	7	3.8–242	11.6+1.43	11.5	unattributed; no recurrence	
HD 14055	X1190d88_X4526	7	344.4676	344.4515	5.006	— SO(8 ₈ –7 ₇)	+192.7	30	20–611	11.6+2.24	12.2	unattributed; no recurrence	
the identical channel of two different spectral windows	Xf8f6a9_X10359	6	218.8729	218.8923	5.090	5.56 H ₂ CO(3 _{2,1} –2 _{2,0})	+181.2	7	18–126	10.5+1.52	11.5	unattributed; no recurrence	

which is the signature of a fixed intermediate frequency; and the block's 24 sibling executions of the same field, with the same array and the same spectral setup, produce no crossing at all. It is *not* a terrestrial carrier at a fixed sky frequency: each of the four frequencies is covered by 127–137 further windows toward 17–22 other stars, and 0 of them carries a crossing in the same channel. **The evidence supports a data-quality failure of one execution block, and that is how these four crossings are positioned** (§5.3.2): under a rule fixed before it was applied — a window is flagged when the median of its own control ensemble exceeds twice the survey median of that quantity — 5 of the 1651 windows are flagged, toward 2 stars (BD+05 1668 and HD 48370), the remain-

ing one being a resolved disc rather than a defect. The flag is a statement about the window, not an attribution of the feature.

None of this is a substitute for the ON/OFF cadence a dedicated search uses, and we do not present it as one. Accordingly we claim only that *no identified interference mechanism accounts for the unattributed events*: allocation tables cannot exclude harmonics, intermodulation products, unconventional emitters or near-field sources, and an archival dataset without a simultaneous off-source strategy cannot do better. What the four tests establish is that the crossing population as a whole carries none of the signatures interference leaves — fixed sky frequency, fixed intermediate frequency, or presence in unrelated targets — which is a statement about the population and not a clearance of any individual crossing.

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